



南京大學
NANJING UNIVERSITY

互联网计算

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智能软件与工程学院

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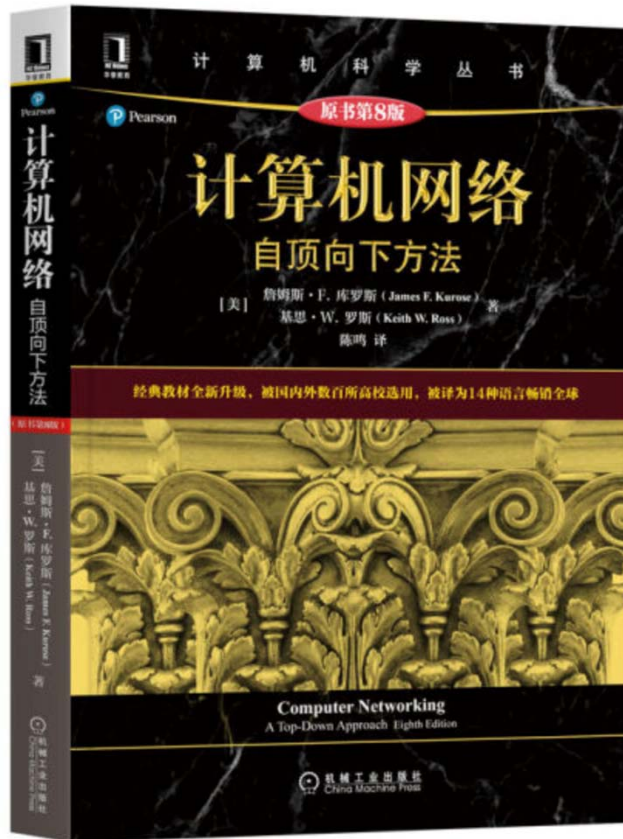
教学内容

- 计算机网络和因特网（2学时）
- 应用层（2学时）
- 运输层（6学时）
- 网络层：数据平面（4学时）
- 网络层：控制平面（4学时）
- 链路层和局域网（6学时）
- 无线网络和移动网络（2学时）
- 计算机网络中的安全（4学时）
- 习题课和总复习（2学时）

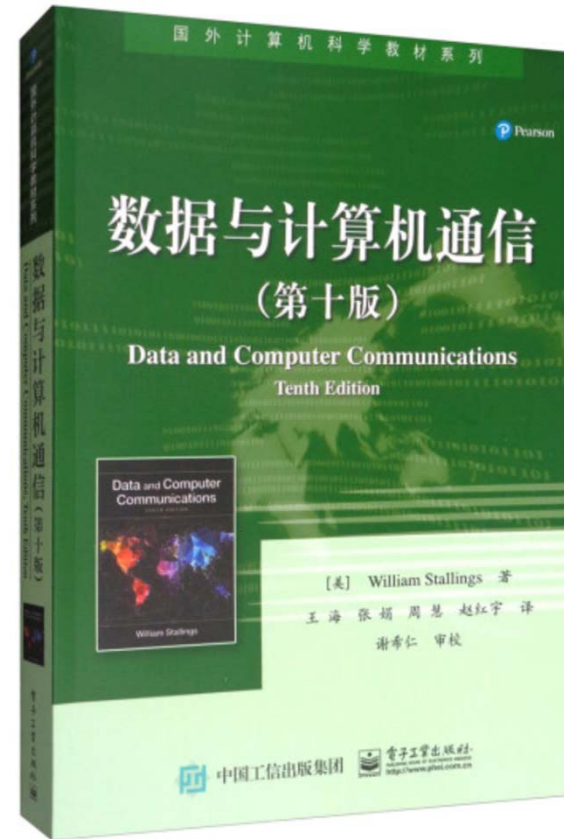


课程书籍

- 教材



- 参考文献





考核形式（暂定）

- **平时成绩（包括上课、课后作业）：30%**
 - 随机抽查上课情况
 - 约3-4次作业
- **考试成绩：70%**



Computer Networks and Internet

- Basic Concepts and Questions
- Internet History
- Protocol Layers and Service Model
- Network Performance

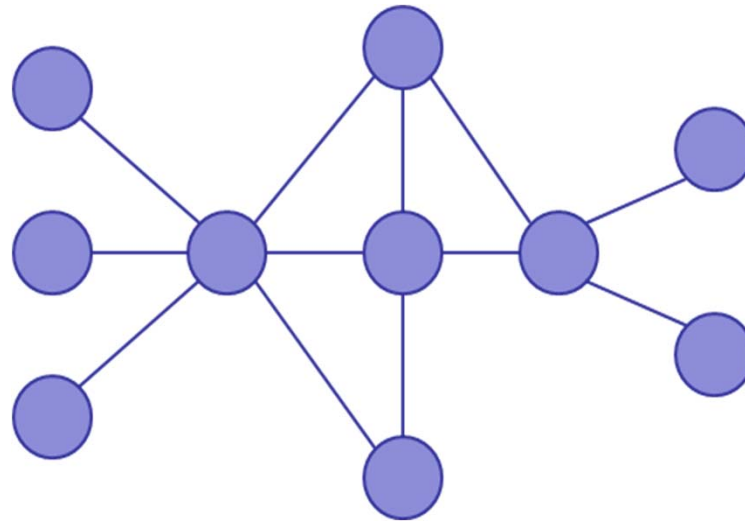


Concept: Internet



What is a network?

- A system of “links” that interconnect “nodes” in order to move “information” between nodes



- Yes, this is very vague



Different types of networks

- Internet
- Telephone network
- Wireless networks
- Optical networks
- Datacenter networks
- Transportation networks
- Social networks

We will focus primarily on the Internet.





What is the Internet?

- [WiKi]
 - The Internet is the **global system** of **interconnected** mainframe, personal, and wireless computer networks that use the **Internet protocol suite** (TCP/IP) to link billions of devices worldwide.
 - It is a **network of networks** that consists of millions of private, public, academic, business, and government networks of local to global scope, linked by a broad array of electronic, wireless, and optical networking technologies.



The Internet is transforming everything

- **The way we do business**
 - E-commerce, advertising, cloud-computing
- **The way we have relationships**
 - Facebook friends, E-mail, IM, virtual worlds
- **The way we learn**
 - Wikipedia, MOOCs, search engines
- **The way we govern and view law**
 - E-voting, censorship, copyright, cyber-attacks



MASSIVE Scale

- 3.5 Billion users (34% of world population)
- 1 Trillion websites
- 200 Billion emails sent per day
- 2 Billion smartphones
- 1.8 Billion Facebook users
- 4 Billion YouTube videos watched per day
- Routers that switch 10 Terabits/second
- Links that carry 100 Gigabits/second



Diversity in all dimensions

- **Technology**
 - Optical, wireless, satellite, copper
- **Endpoint devices**
 - From wearable devices and cell phones to datacenters and supercomputers
- **Applications**
 - Video streaming, social networking, file transfer, Skype, live TV, gaming, remote medicine, IM
- **Users**
 - Malicious, naïve, savvy, embarrassed, paranoid



The Internet is a lesson

- In how to reason through the design of a very complex system
 - What are our goals and constraints?
 - What's the right prioritization of goals?
 - How do we decompose a problem?
 - Who does what? How?
 - What are the interfaces between components?
 - What are the tradeoffs between design options?



Basic questions:

Q1: What is the Internet made of?

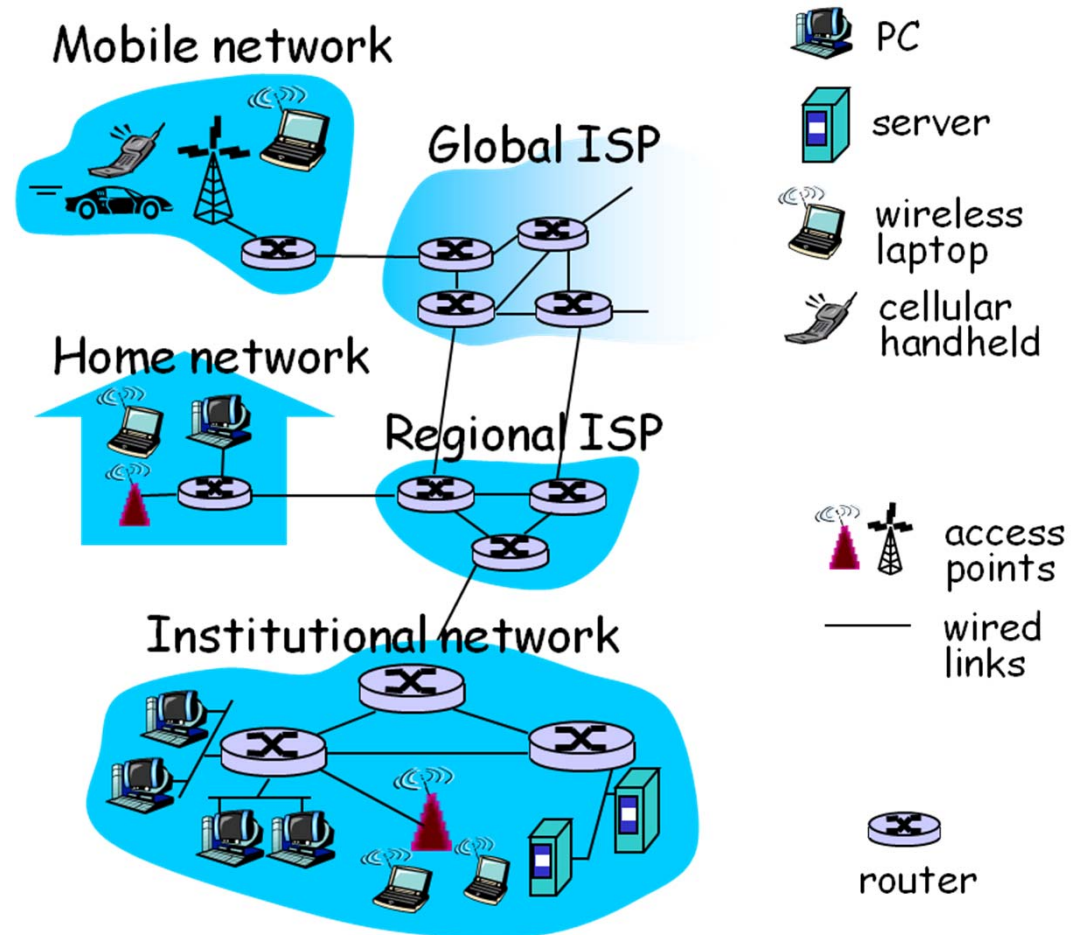
Q2: How to connect to the Internet?

Q3: How to transfer data in the network?



Internet - Component View

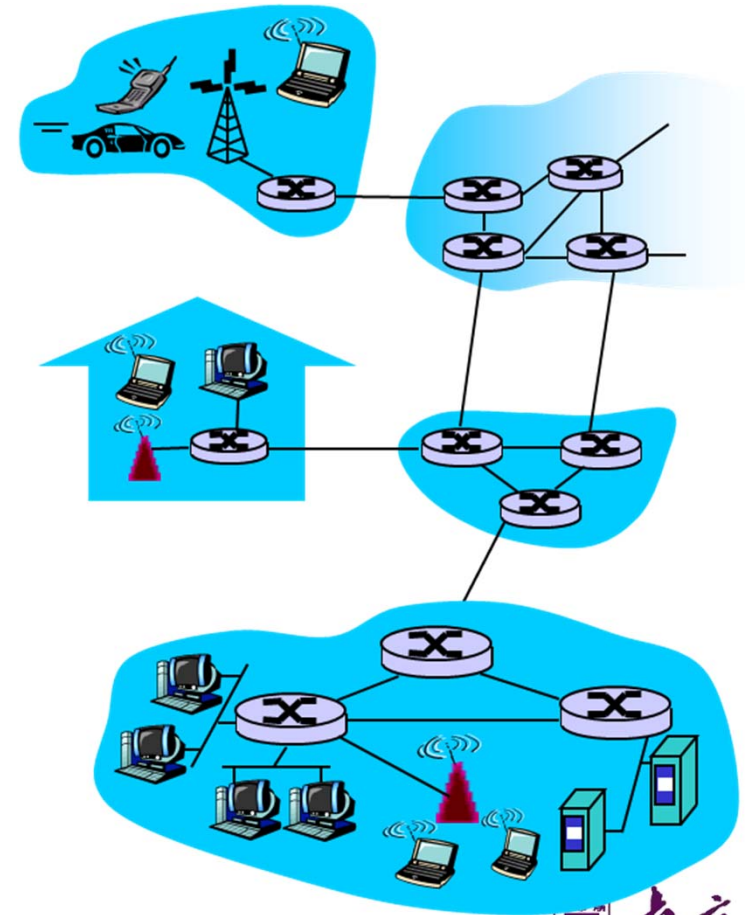
- Millions of connected **computing devices**
 - Hosts = **End systems**
 - Running network applications
- **Communication links**
 - Fiber, Copper, Radio, Satellite
 - Building physical networks
- **Routers**
 - Forward packets (chunks of data) between physical networks





Internet - Service View

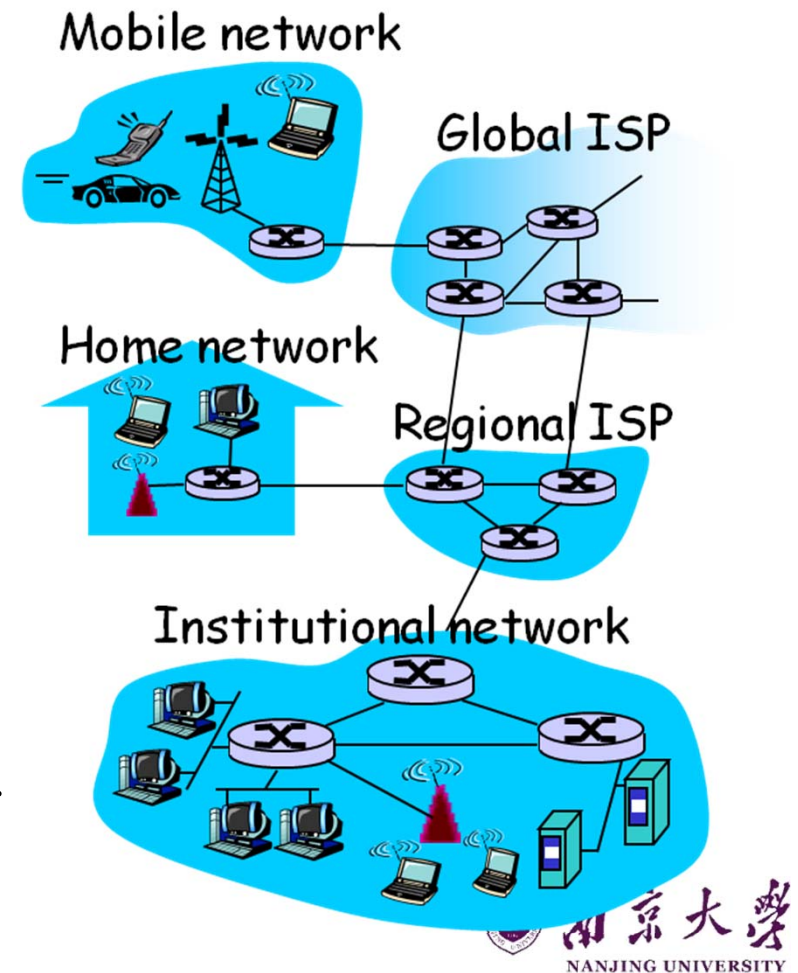
- Communication infrastructure
 - Enables distributed applications
 - Web, VoIP, email, online games, e-commerce, file sharing
- Communication services provided to Apps:
 - Reliable data delivery from source to destination
 - “best effort” (unreliable) data delivery
 - Guaranteed delay and throughput





Internet - Protocols

- Network Protocols
 - Control sending, receiving of messages
 - e.g. HTTP, Skype; TCP, IP; PPP, Ethernet
- Internet standards
 - IETF: Internet Engineering Task Force
 - RFC: Request for comments
- Internet: "network of networks"
 - Public Internet versus private Intranet
 - Loosely hierarchical





What's a protocol?

human protocols:

- “what's the time?”
 - “I have a question”
- ... specific msgs sent
- ... specific actions taken when msgs received, or other events

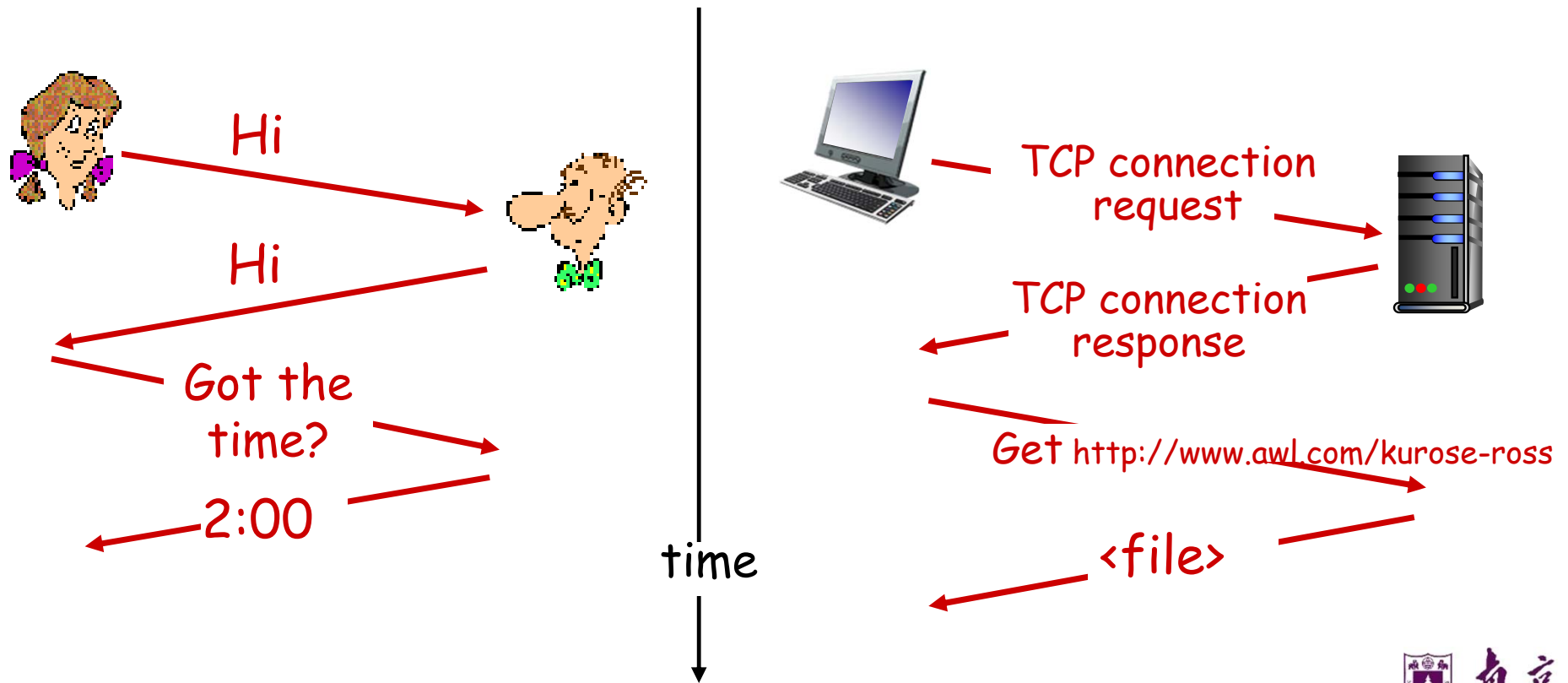
network protocols:

- machines rather than humans
- all communication activity in Internet governed by protocols

protocols define **format**, **order** of **msgs sent and received** among network entities, and **actions taken** on msg transmission, receipt



a human protocol and a computer network protocol:



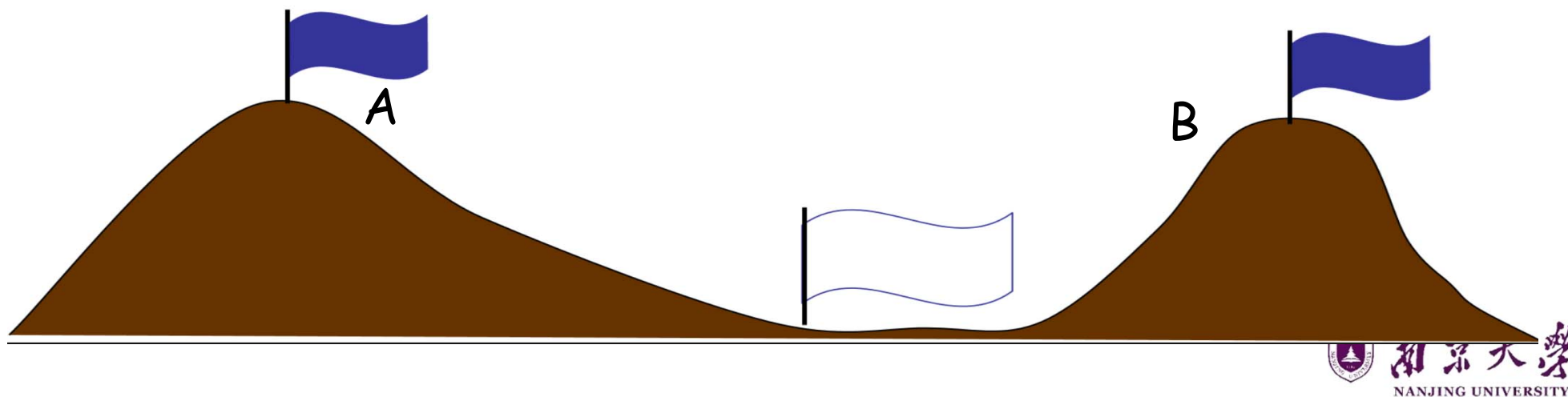


Reliability of Communication

- 例子：蓝军-白军作战

占据东、西两个山顶的蓝军A和蓝军B与驻扎在山谷的白军作战。其力量对比是：单独的蓝军A或蓝军B打不过白军，但蓝军A和蓝军B协同作战则可战胜白军。

- A和B可以派遣通信兵穿过白军营地向对方发送消息，但是通信兵有可能被白军截获。
- 现蓝军A拟于次日正午向白军发起攻击。有什么方法能保证蓝军取得胜利？





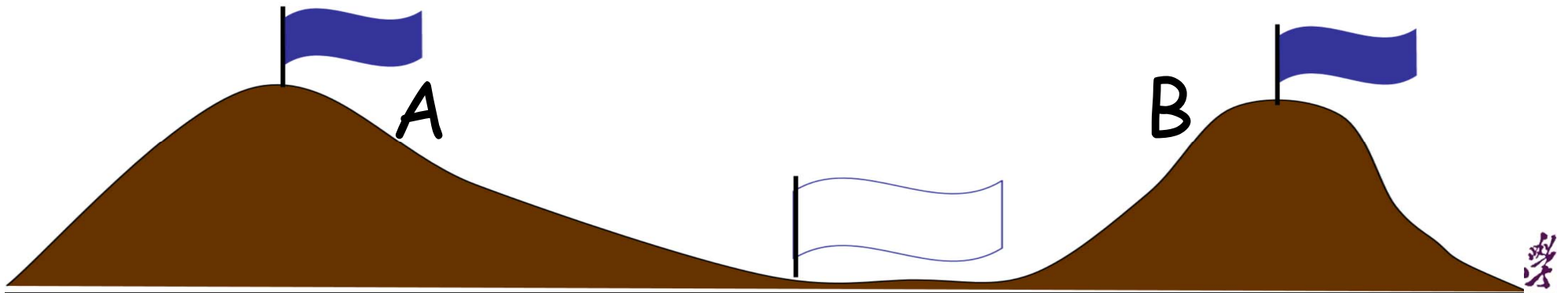
明日正午进攻，如何？

收到“同意”

同意

收到：收到“同意”

- 这样的协议无法实现！
没有方法能保证蓝军100%取得胜利





Basic questions:

Q1: What is the Internet made of?

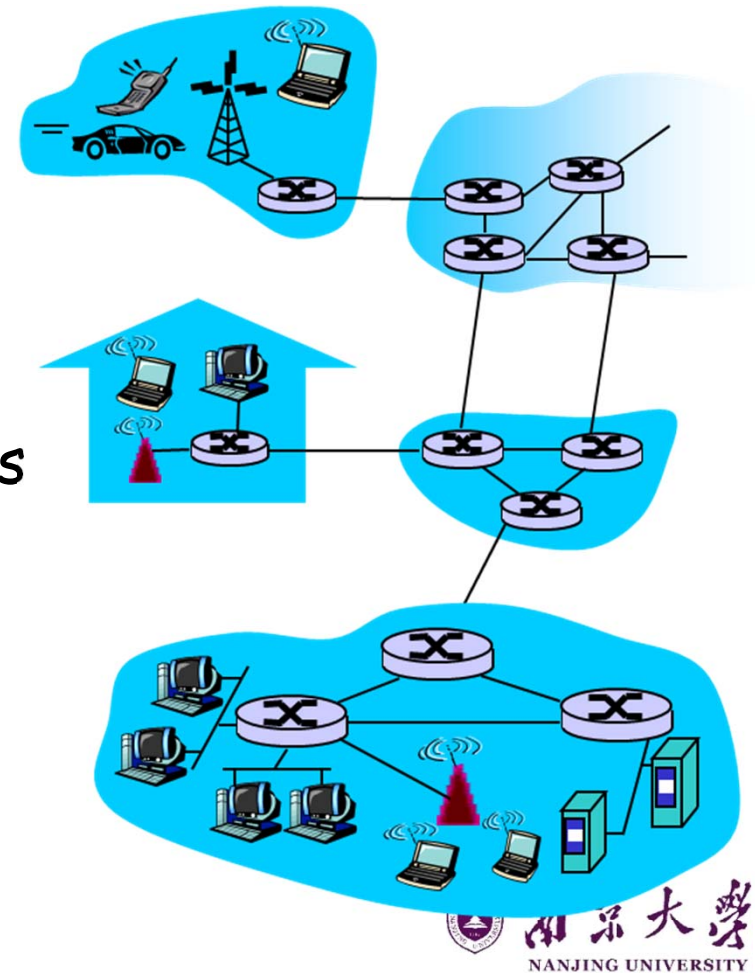
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Access Internet

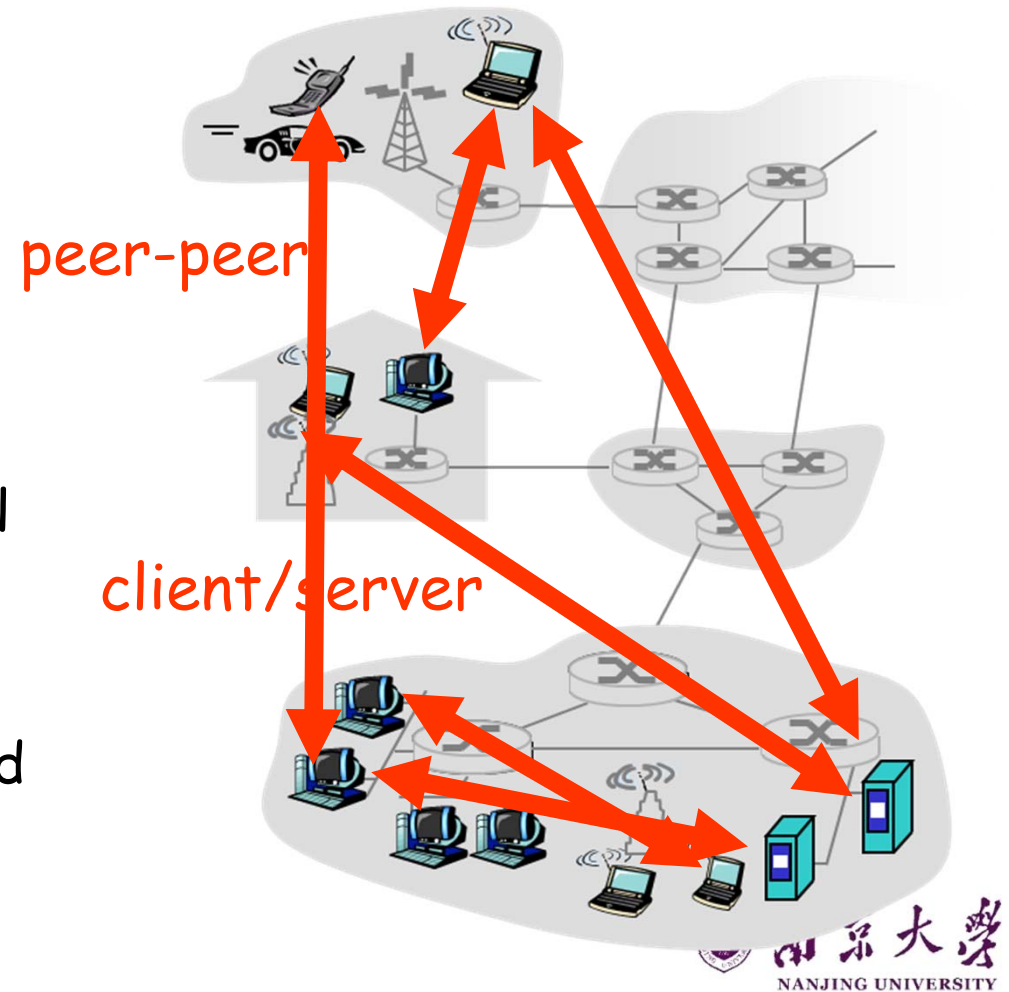
- Network edge
 - Applications and hosts
- Access networks
 - Physical media
 - Wired and wireless communication links
- Network core
 - Interconnected routers
 - Network of networks





Network Edge

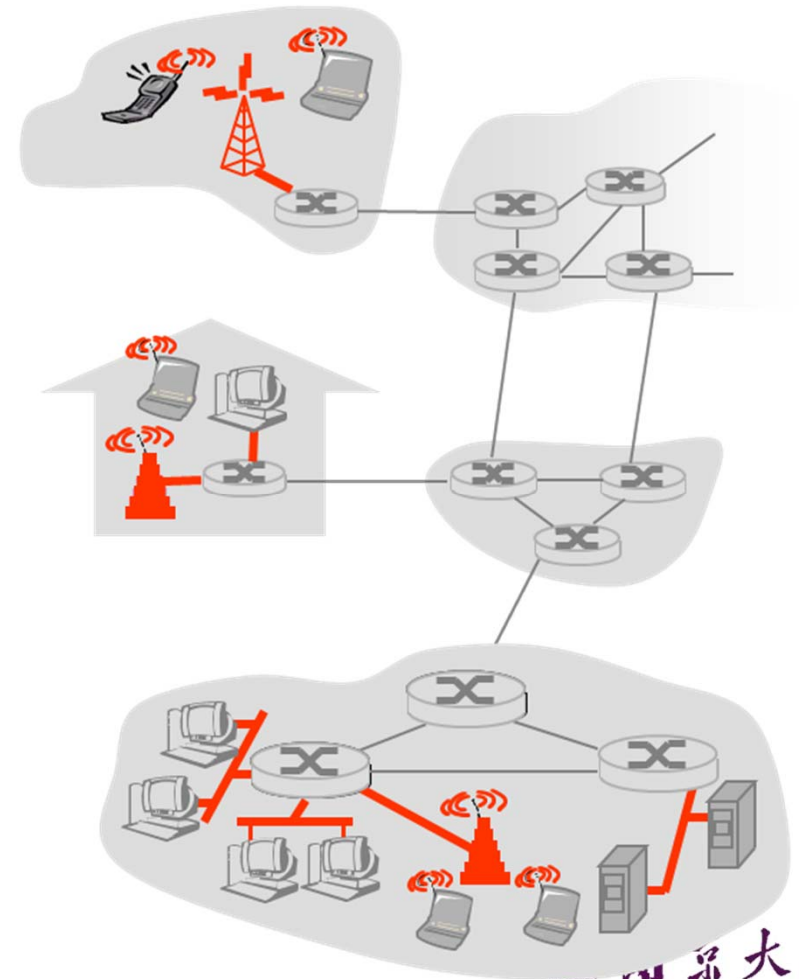
- End systems (hosts)
 - Run application programs
 - e.g. Web, Email
- Client/server model
 - Client host requests, receives service from always-on server
 - e.g. Web browser/server; Email client/server
- Peer-to-peer model
 - Minimal (or no) use of dedicated servers
 - e.g. Skype, BitTorrent





Access Networks

- How to connect **end systems to edge router?**
 - Residential (Home) access networks
 - Institutional access networks (school, company)
 - Mobile access networks





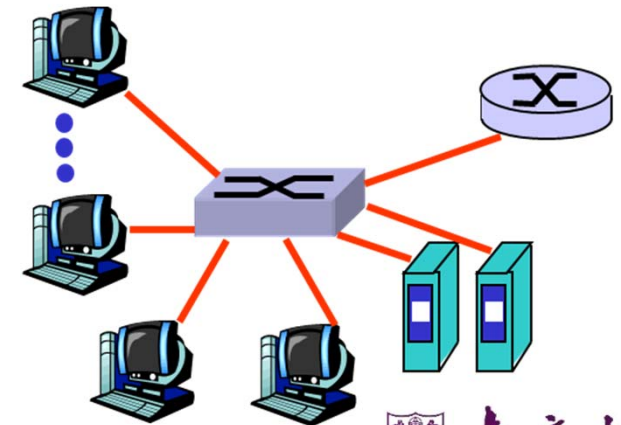
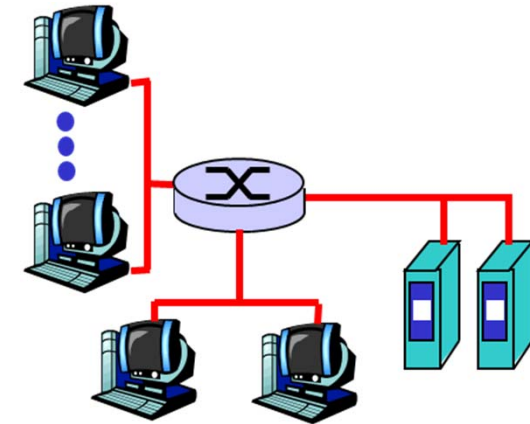
Residential Access

- **Dialup via modem**
 - Up to 56Kbps direct access to router
- **DSL: digital subscriber line**
 - Deployment: telephone company
 - Up to 1 Mbps upstream, and 8 Mbps downstream
 - **Dedicated** physical line to telephone central office
- **HFC: hybrid fiber coax**
 - Asymmetric: up to 30Mbps downstream, 2 Mbps upstream
 - **Homes share** access to ISP router
 - Deployment: cable TV companies



Company Access: Local Area Networks

- Company/University **local area network** (LAN) connects end systems to edge router
- **Ethernet**:
 - 10 Mbps, 100Mbps, 1Gbps, 10Gbps Ethernet
 - Modern configuration: end systems connect into backbone of Ethernet switches





Wireless Access Networks

- Shared wireless media connects end system to router
 - via base station, or “access point”
- Wireless LANs:
 - 802.11b/g (WiFi): 11 or 54 Mbps
- Wide-area wireless access
 - Provided by telecommunication operator, 10's Km
 - between 1 and 10 Mbps
 - 3G, 4G: LTE, WiMax



to Internet



to Internet



Physical media

- **bit**: propagates between transmitter/receiver pairs
- **physical link**: what lies between transmitter & receiver
- **guided media**:
 - signals propagate in solid media: copper, fiber, coax
- **unguided media**:
 - signals propagate freely, e.g., radio
- **twisted pair (TP)**
 - two insulated copper wires
 - Category 5: 100 Mbps, 1 Gbps Ethernet
 - Category 6: 10Gbps

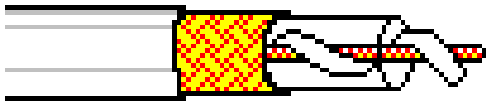




Physical media: coax, fiber

coaxial cable:

- two concentric copper conductors
- bidirectional
- broadband:
 - multiple channels on cable



fiber optic cable:

- ❖ glass fiber carrying light pulses, each pulse a bit
- ❖ high-speed operation:
 - high-speed point-to-point transmission (e.g., 10's-100's Gpbs transmission rate)
- ❖ low error rate:
 - repeaters spaced far apart
 - immune to electromagnetic noise





Physical media: radio

- signal carried in electromagnetic spectrum
- no physical “wire”
- bidirectional
- propagation environment effects:
 - reflection
 - obstruction by objects
 - interference

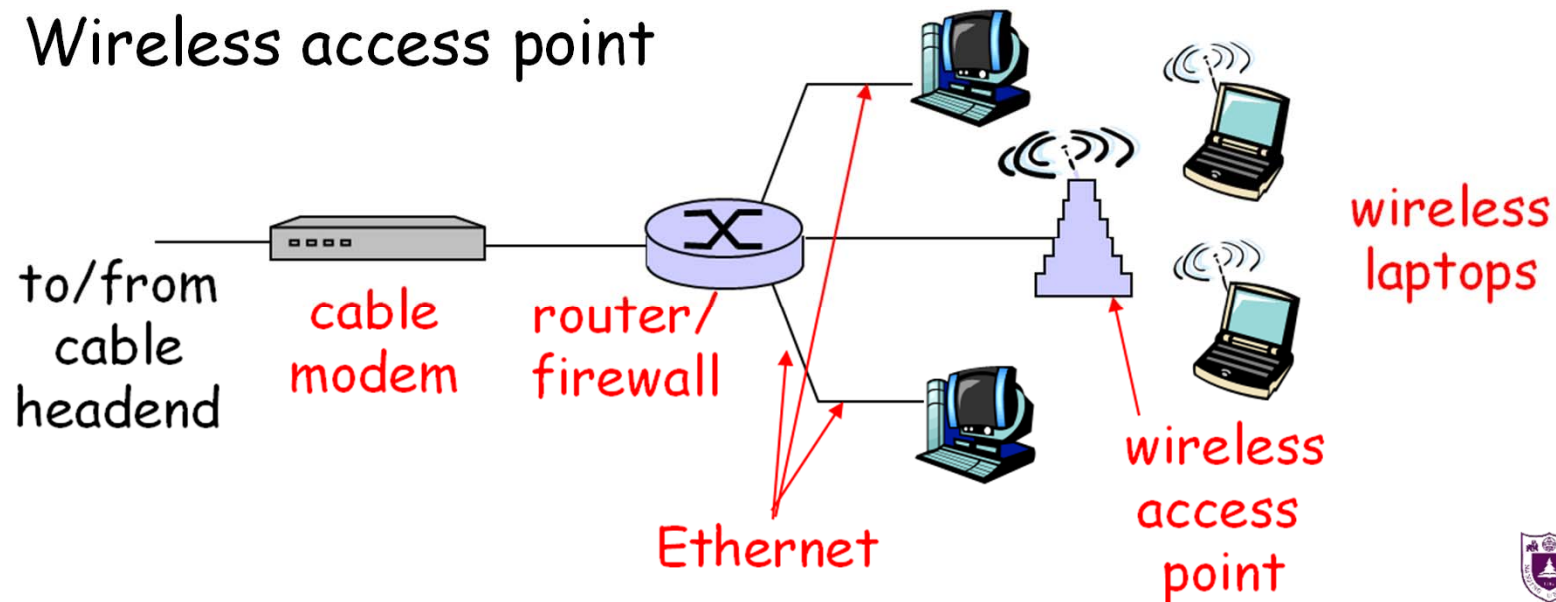
radio link types:

- ❖ **terrestrial microwave**
 - e.g. up to 45 Mbps channels
- ❖ **LAN** (e.g., WiFi)
 - 11Mbps, 54 Mbps
- ❖ **wide-area** (e.g., cellular)
 - 3G cellular: ~ 1 Mbps
- ❖ **satellite**
 - Kbps to 45Mbps channel (or multiple smaller channels)
 - 270 msec end-end delay
 - geosynchronous versus low altitude



Example: A Modern Family

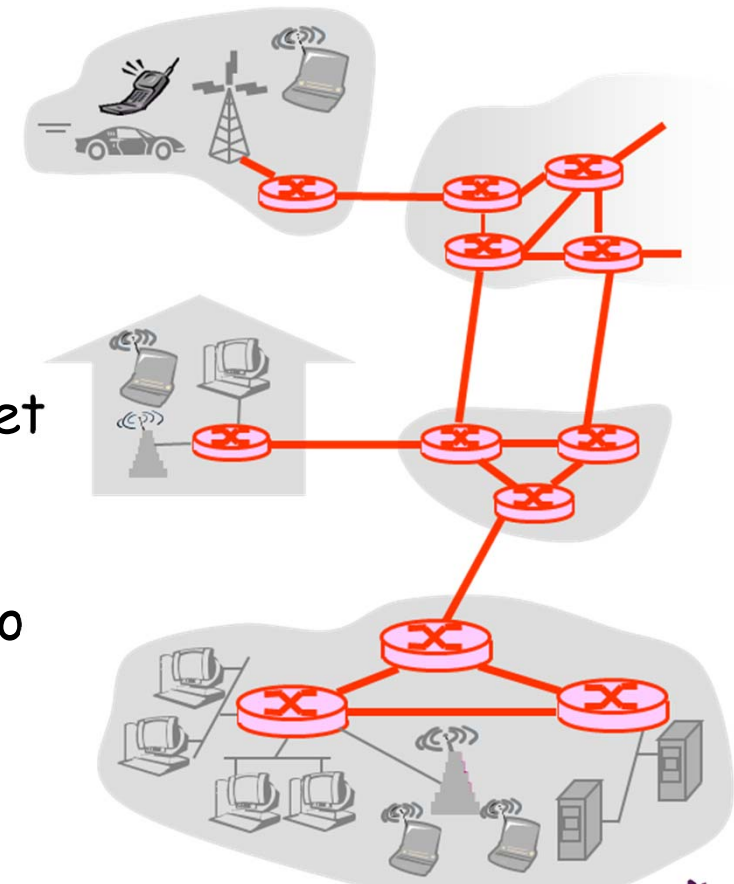
- A **home network** components:
 - DSL or cable modem
 - Router/Firewall/NAT
 - Ethernet switch
 - Wireless access point





The Network Core

- Mesh of **interconnected routers**
- **Fundamental question**
 - How is data transferred through the net?
- **Circuit switching**
 - Dedicated circuit per call, e.g. telephone net
- **Packet-switching**
 - hosts break application-layer messages into packets
 - forward packets from one router to the next, across links on path from source to destination





- 请归类：

- 属于网络边缘的是：_____
- 属于接入网络的是：_____
- 属于网络核心的是：_____

A 笔记本电脑； B 手机； C 路由器；
D 双绞线； E 智能家具； F 无线路由器；
G 服务器； H 同轴电缆； I 交换机



- 请归类:

- 属于网络边缘的是: A B E G
- 属于接入网络的是: D F H
- 属于网络核心的是: C I

A 笔记本电脑; B 手机; C 路由器;
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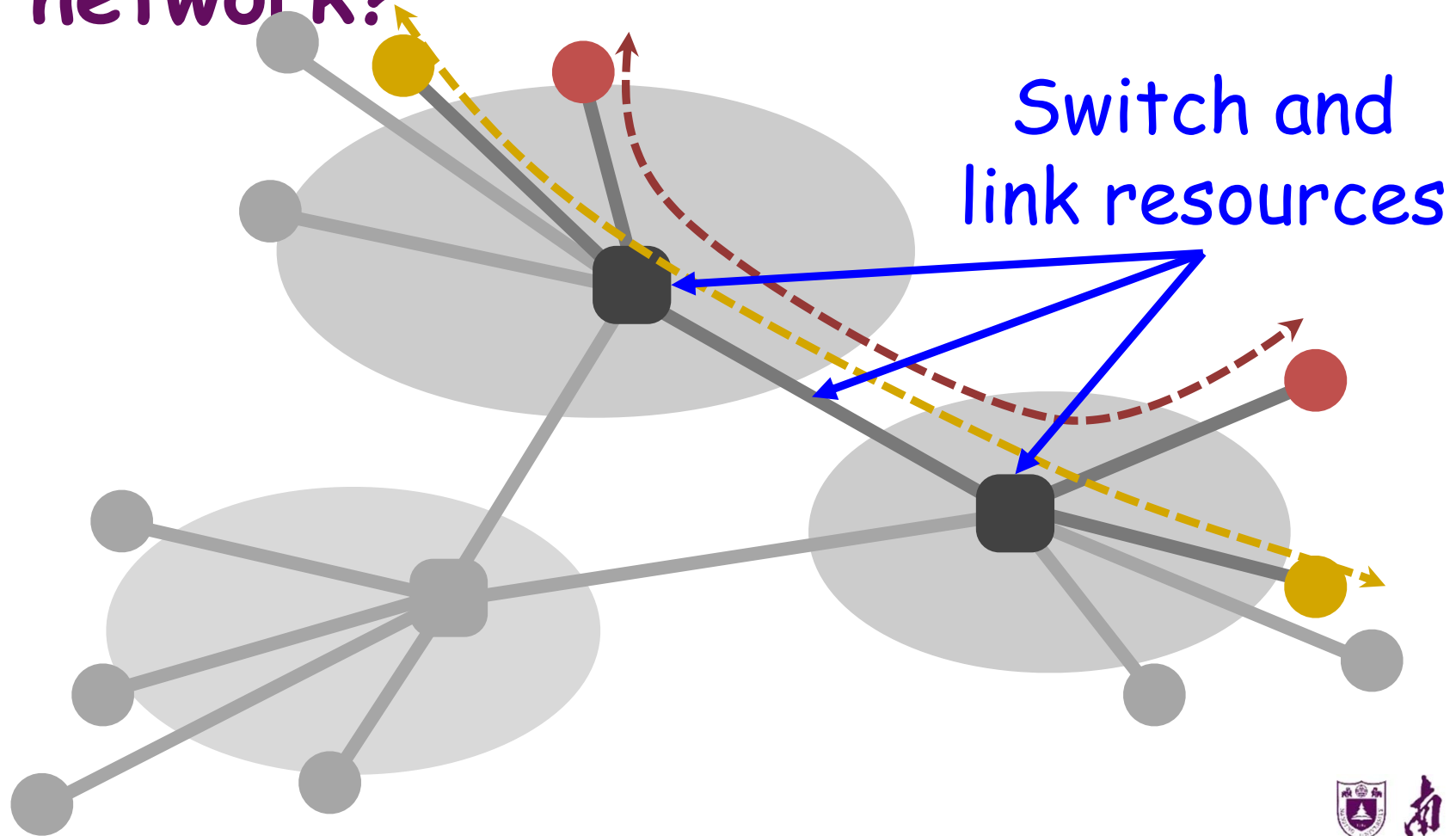


Switched networks

- End-systems and networks connected by switches instead of directly connecting them
- Allows us to scale
 - For example, directly connecting N nodes to each other would require N^2 links!



When do we need to share the network?





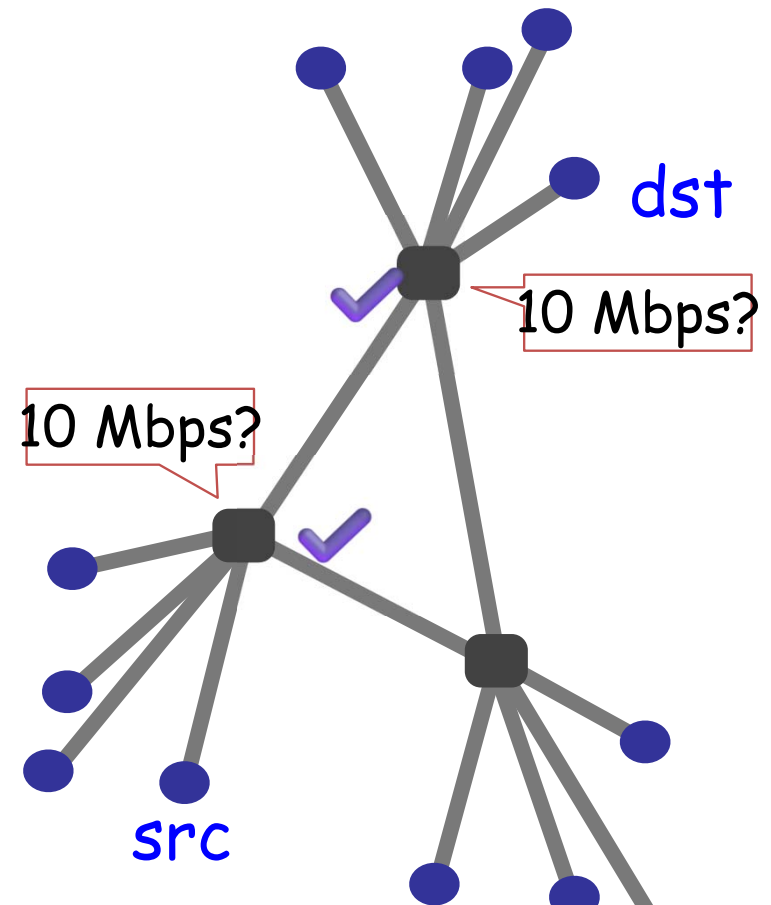
Two ways to share switched networks

- **Circuit switching**
 - Resource reserved per connection
 - Admission control: per connection
- **Packet switching**
 - Packets treated independently, on-demand
 - Admission control: per packet
- **Hybrid: virtual circuits**
 - Emulating circuit switching with packets (see text)



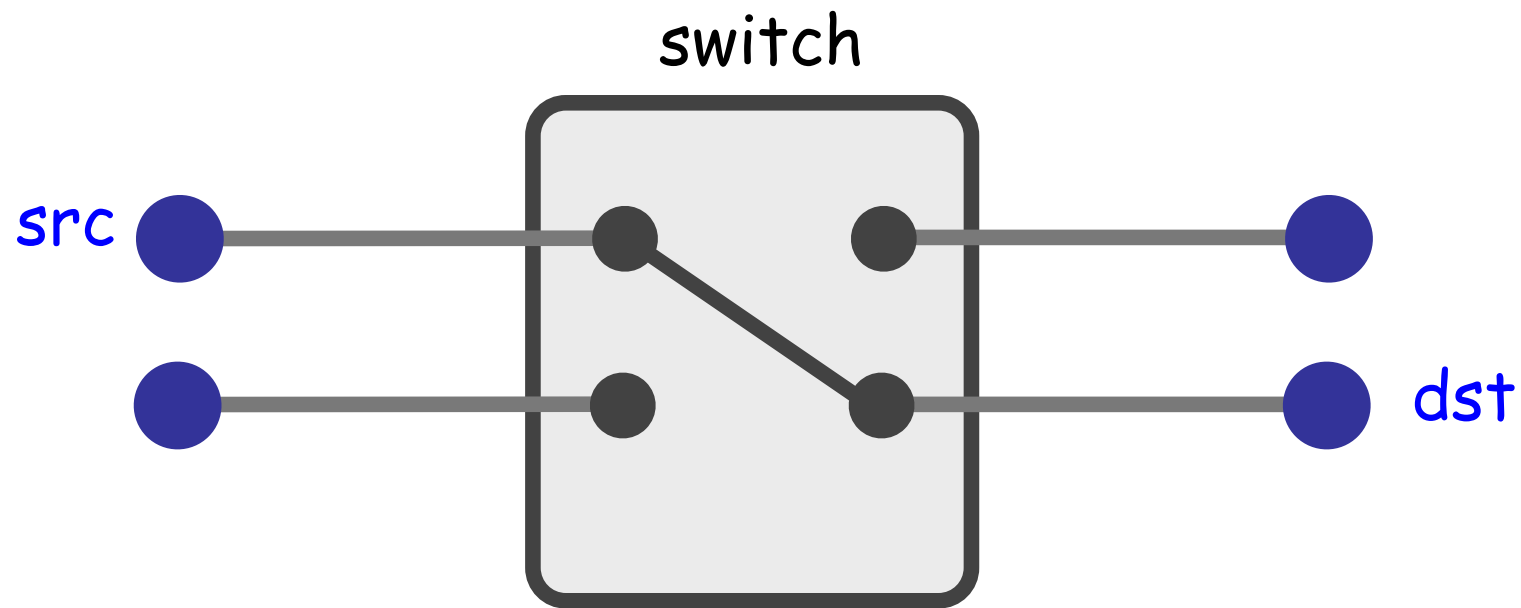
Circuit switching

1. **src** sends reservation request to **dst**
2. Switches create circuit after admission control
3. **src** sends data
4. **src** sends teardown request





Circuit switching



- Reservation establishes a "circuit" within a switch

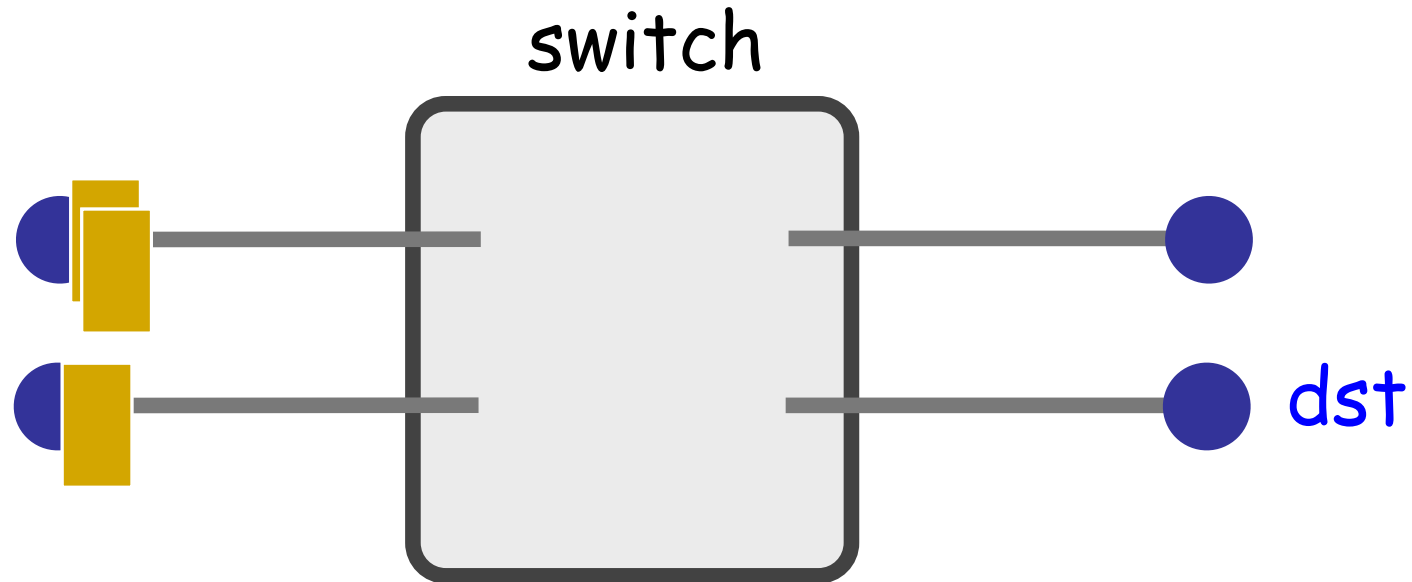


Circuit switching

- Pros
 - Predictable performance
 - Simple/fast switching (once circuit established)
- Cons
 - Complexity of circuit setup/teardown
 - Dedicated resources: Inefficient when traffic is bursty
 - Circuit setup adds delay
 - Switch fails → its circuit(s) fails



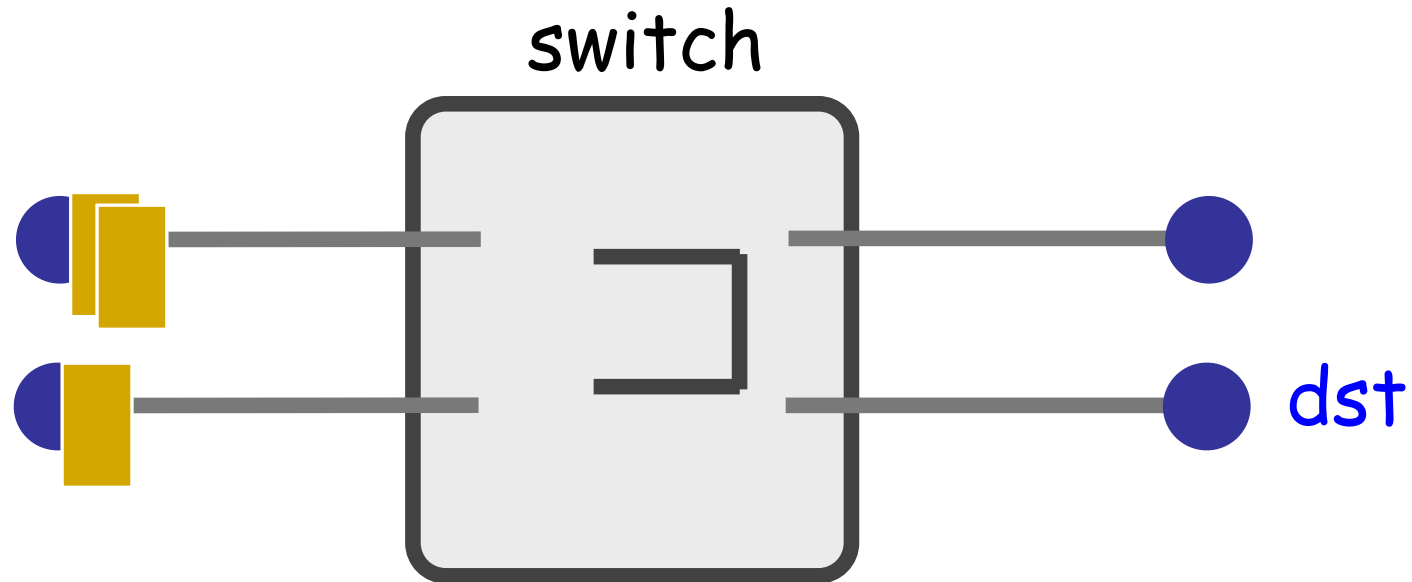
Packet switching



- Each packet contains destination (**dst**)
- Each packet treated independently



Packet switching

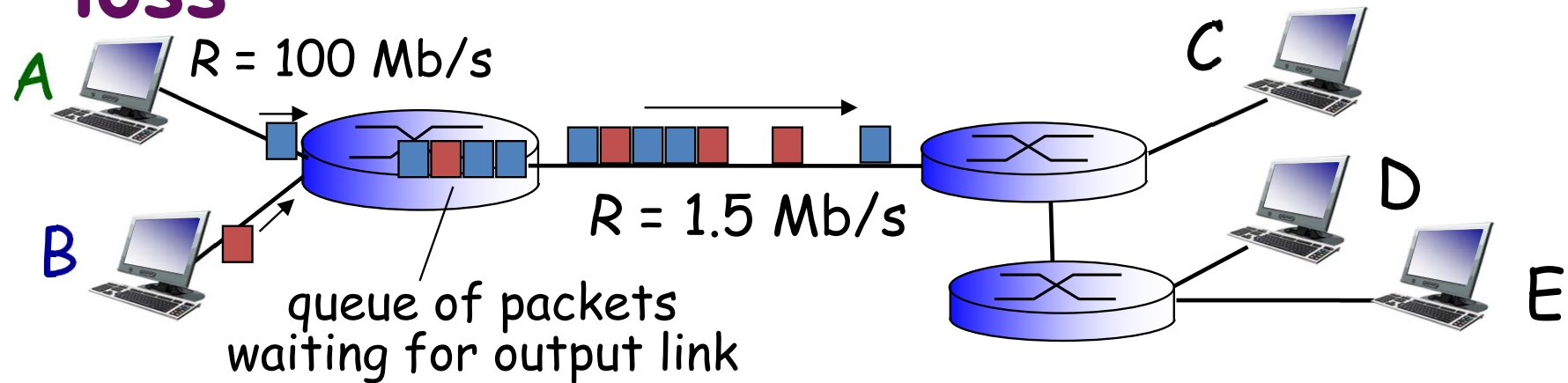


- Each packet contains destination (**dst**)
- Each packet treated independently
- With buffers to absorb transient overloads

Store and forward: packets move one hop at a time, stored (queued) at switches



Packet Switching: queueing delay, loss



- Resource contention
 - aggregate (burst-up) resource demand can exceed amount available
- Congestion:
 - packets will queue, wait for link use
 - packets can be dropped (lost) if no memory to store them



Packet Switching

- Pros
 - Efficient use of network resources
 - Simpler to implement
 - Robust: can "route around trouble"
- Cons
 - Unpredictable performance
 - Requires buffer management and congestion control



Statistical multiplexing

- Statistical Multiplexing (统计多路复用):
Link bandwidth shared on demand (按需共享)
- Allowing more demands than the network can handle
 - Hoping that not all demands are required at the same time
 - Results in unpredictability
 - Works well except for the extreme cases



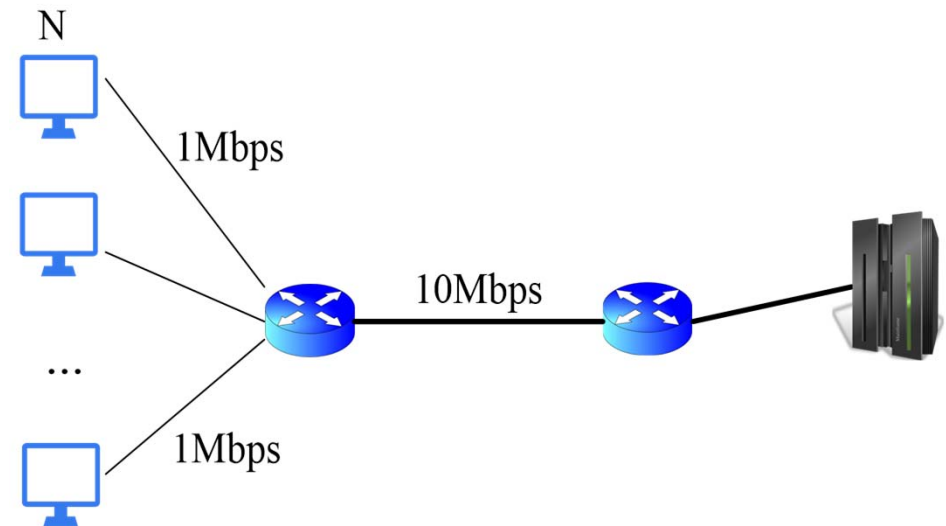
Example: Statistical Multiplexing

Example:

- N users share one link (10Mbps)
 - Each user requires 1Mbps
 - Each user: active 10%, idle 90%.
- How many users are supported?

Circuit Switching:

$N = 10\text{Mbps} / 1\text{Mbps} = 10$ users



Statistical Multiplexing:

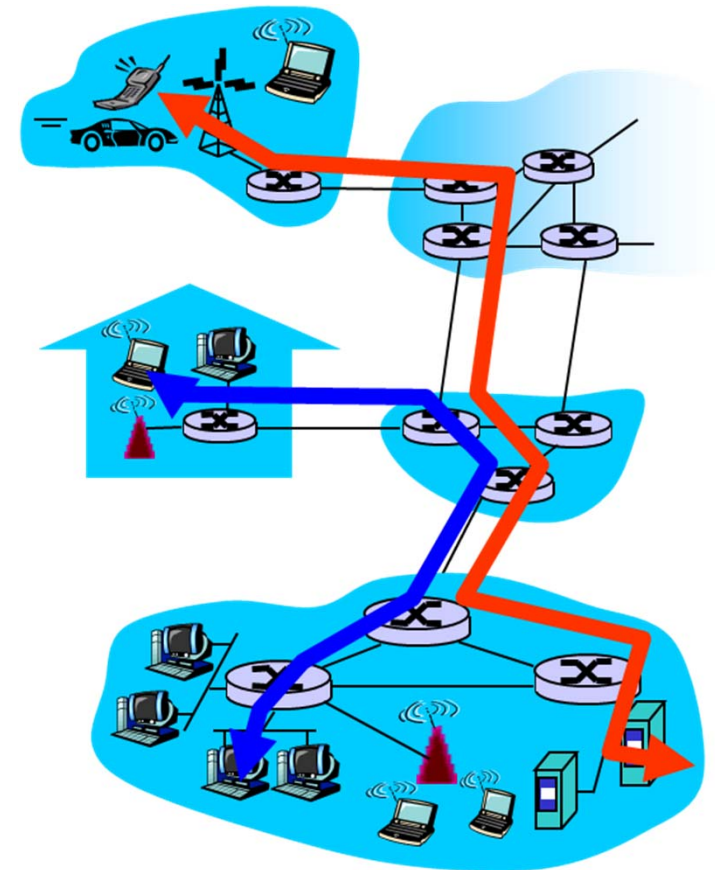
Assume $N=35$, $\text{Prob}\{\text{active user} > 10\} \leq 0.0004$,

So for $N=35$, with probability 0.9996 a user have bandwidth larger than 1Mbps.



Virtual Circuit

- Circuit Switching + Packet Switching
 - Routes or main cross roads are fixed
 - Resources shared, congestion control needed
 - Resources can be preserved, leading to different performance
 - Connection setup/teardown needed





Comparison

	电路交换	数据报分组交换	虚电路分组交换
传输通路	专用	非专用	非专用
连续性	连续传输	分组传输	分组传输
带宽	固定	动态使用	动态使用
路由	固定	动态	固定
时延	实时（只有呼叫建立时延）	分组传输时延	分组传输时延+呼叫建立时延
扩展性	差（接入用户有上限）	好（用户数量可动态扩充）	较好（用户数量动态，由拥塞控制来保证服务质量）



Computer Networks and Internet

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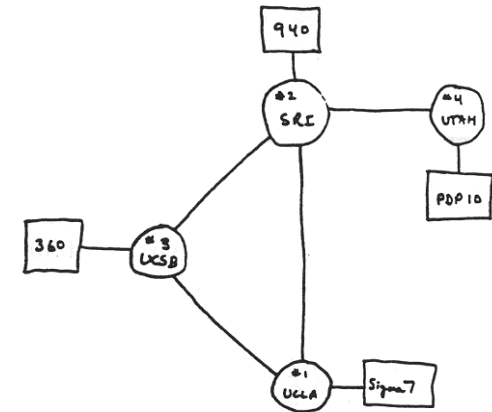


Internet History (1)

1961-1972: Early packet-switching principles

60年代：诞生-分组交换网络

- **1961:** Kleinrock - queuing theory shows effectiveness of packet-switching (PhD@MIT)
- **1964:** Baran - packet-switching in military nets
- **1967:** ARPAnet conceived by Advanced Research Projects Agency
- **1969:** first ARPAnet node operational (UCLA, Stanford, UCSB, UTAH), Kleinrock
- **1972:**
 - ARPAnet demonstrated publicly
 - NCP (Network Control Protocol) first host-host protocol [RFC001]
 - First email program
 - ARPAnet has 15 nodes



THE ARPA NETWORK



In the Press About Publications History Twitter Students

The Day the Infant Internet Uttered its First Words

Below is a record of the first message ever sent over the ARPANET. It took place at 22:30 hours on October 29, 1969. This record is an excerpt from the "IMP Log" that was kept at UCLA. Professor Kleinrock, gma 7 Host computer to another programmer, Bill Duvall, at the SRI SDS 940 mission itself was simply to "login" to SRI from UCLA. They succeeded in he "o" and then the system crashed! Hence, the first message on the Internet shold! They were able to do the full login about an hour later.

Leonard Kleinrock

Born June 13, 1934 (age 82)
New York City
Residence Los Angeles
Nationality United States
Fields Engineering
Computer science
Institutions UCLA



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Internet History (2)

1972-1980: Internetworking, new and proprietary nets

70年代：成型 单一、封闭网络 -> 开放互联网络

- 1970: ALOHAnet satellite network in Hawaii, Norman Abramson (无线分组网络)
- 1973: Robert Metcalfe's PhD thesis (@Harvard) proposes Ethernet (以太网), at Xerox PARC in 1976 (局域网诞生)
- 1974: Cerf and Kahn - architecture for interconnecting networks (Internet构架)
- Late70's:
 - Proprietary architectures: DECnet, SNA, XNA
 - Switching fixed length packets (ATM precursor)
- 1979: ARPAnet has 200 nodes

Vint Cerf, Robert E. Kahn and George W. Bush



- Cerf and Kahn's internetworking principles:
 - Minimalism, autonomy - no internal changes required to interconnect networks
 - Best effort service model
 - Stateless routers
 - Decentralized control
- Define today's Internet architecture
- Design of TCP/IP suits



Internet History (3)

1980-1990: new protocols, a proliferation of networks

80年代：持续发展

- 新协议：NCP-> TCP/IP
 - DNS：实现域名解析
 - 应用：Email, Ftp
- 1983: deployment of TCP/IP
 - 1982: SMTP email protocol defined
 - 1983: DNS defined for name-to-IP-address translation
 - 1985: FTP protocol defined
 - 1988: TCP congestion control
 - New national networks: Cset, BITnet, NSFnet, Minitel
 - 100,000 hosts connected to confederation of networks



Internet History (4)

1990's, 2000's: commercialization, the Web, new apps

90年代：因特网爆炸

- 万维网出现：www (http, HTML, Web Server, Browser)
- 商用化，逐渐普及
- 新型应用：Email, Web, IM (instant messaging) , MP3文件共享

- **Early 1990's:** ARPAnet decommissioned
- **1991:** NSF lifts restrictions on commercial use of NSFnet (decommissioned in 1995)
- **Early 1990's:** Web
 - Hypertext [Bush 1945, Nelson 1960's]
 - HTML, HTTP: Berners-Lee
- **1994:** Mosaic, later Netscape Browser
- **Late 1990's:** commercialization of the Web
- **Late 1990's ~ 2000's:**
 - More killer apps: instant messaging, peer2peer file sharing (e.g. Napster)
 - Network security to forefront
 - Est. 50 million host, 100 million+ users
 - Backbone links running at Gbps

蒂姆·伯纳斯-李爵士
Sir Tim Berners-Lee



出生 1955年6月8日 (61歲) ^[1]

✚ 英格兰伦敦

机构 万维网联盟

南安普敦大学

Plessey

麻省理工学院

知名于 发明万维网

麻省理工学院计算机科学与人工智能实

验室创办主席

2016 [Turing Award](#)



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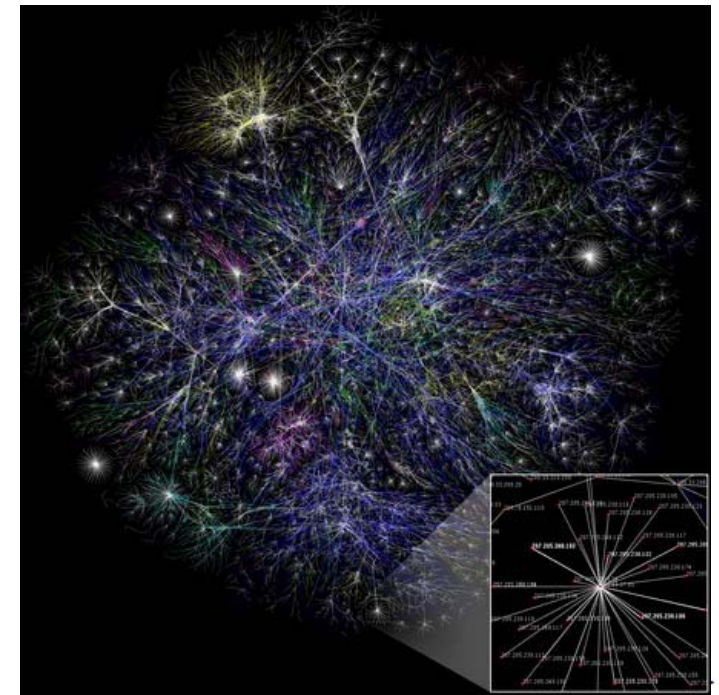
Internet History (5)

2000年以后，新型应用涌现

- 多媒体
- P2P网络
- 社交网络 (Facebook, Twitter, 人人, 微博, 微信, ...)

2007

- ~500 million hosts
- Voice, Video over IP
- **P2P applications**: BitTorrent (file sharing), Skype (VoIP), PPLive (video)
- More applications: YouTube, online gaming
- Wireless and mobility
- 2015- , blockchain, AINet, 5G, ...





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Protocol “layers”

Networks are complex,
with many “pieces”:

- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

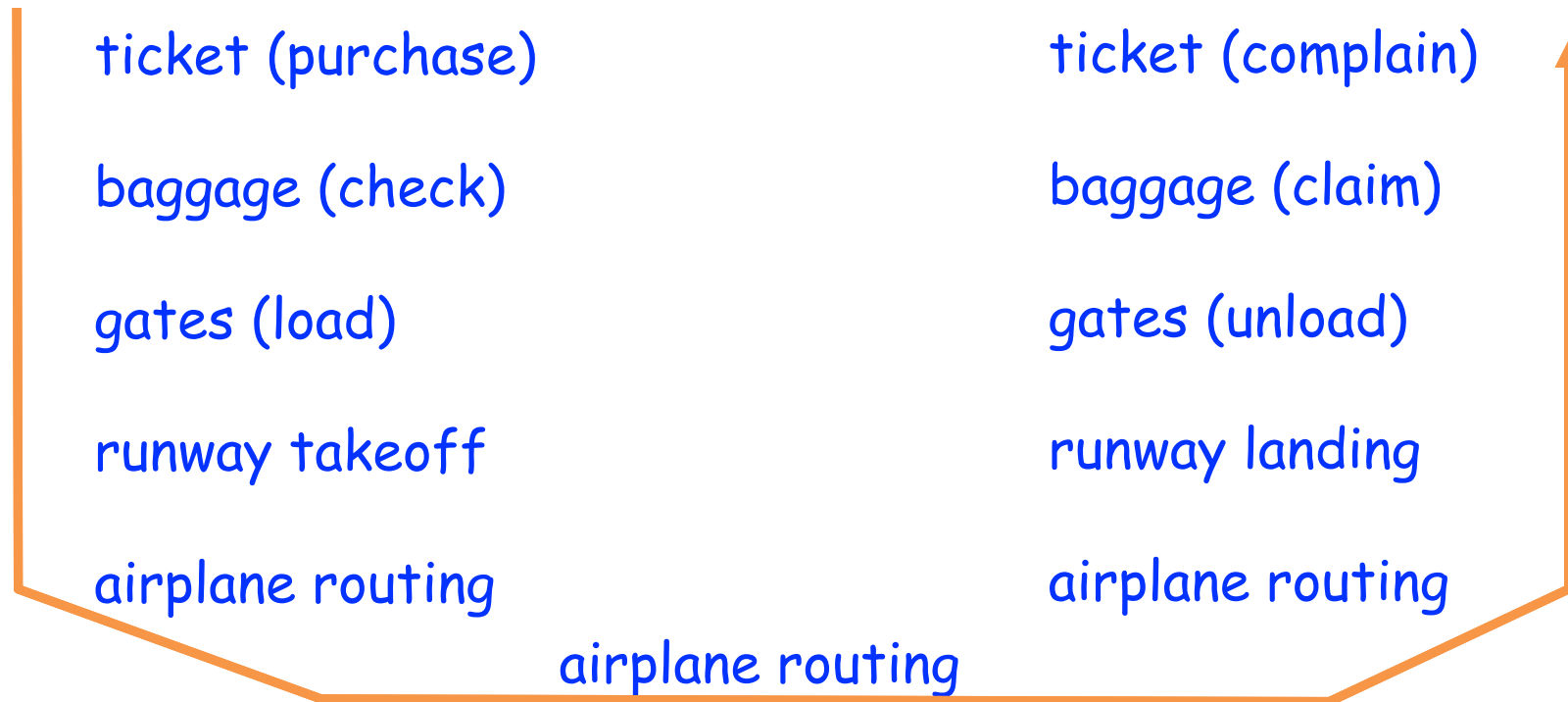
Question:

is there any hope of
organizing structure
of network?

.... or at least our
discussion of
networks?



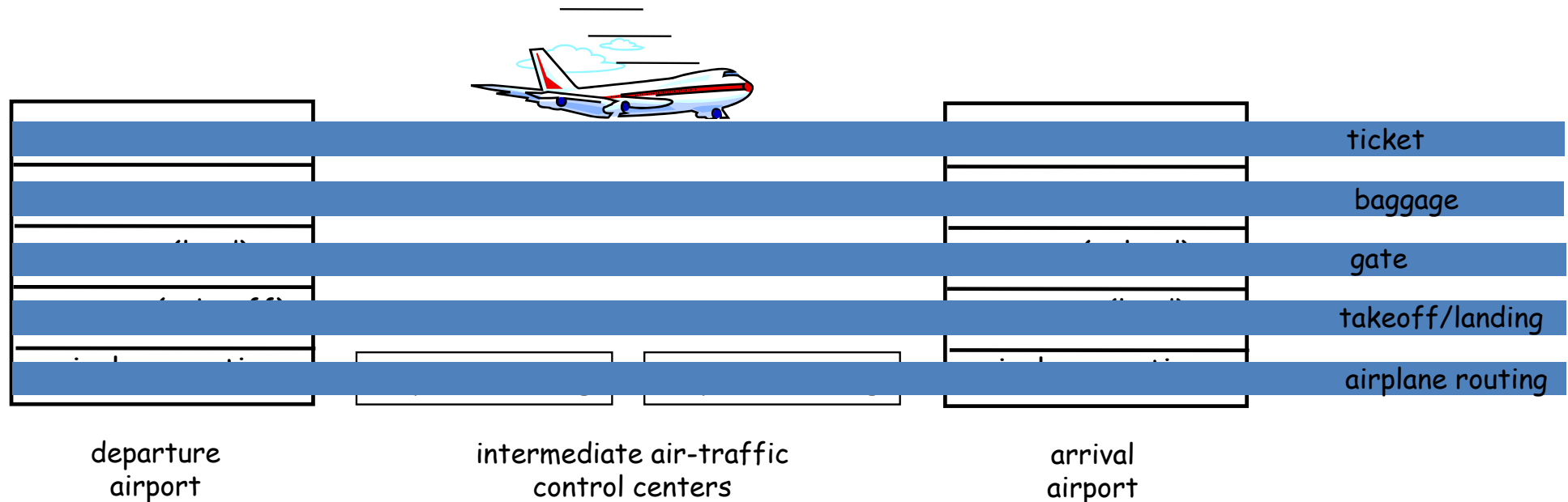
Organization of air travel



- a series of steps



Layering of airline functionality



layers: each layer implements a service

- via its own internal-layer actions
- relying on services provided by layer below



Why layering?

dealing with complex systems:

- explicit structure allows identification, relationship of complex system's pieces
 - layered *reference model* for discussion
- modularization eases maintenance, updating of system
 - change of implementation of layer's service transparent to rest of system
 - e.g., change in gate procedure doesn't affect rest of system



Standard Protocol Architectures

- Two standards:
 - OSI Reference model
 - Never lived up to early promises
 - TCP/IP protocol suite
 - Most widely used
- Others
 - IBM Systems Network Architecture (SNA)
 - DECNet, Netware



ISO-OSI

- Open Systems Interconnection (OSI)
- Developed by the International Organization for Standardization (ISO)
- Seven layers structure
- A theoretical system delivered too late
- TCP/IP is the de facto standard now



OSI - The Model

- A layer model, and flow structure
- Each layer **performs a subset** of the required communication functions
- Each layer **relies on the next lower layer** to perform more primitive functions
- Each layer **provides services** to the next higher layer
- **Changes** in one layer should not require changes in other layers



OSI Layers

Example:
Alice invite Bob to lunch



"请客吃饭"

语言表述

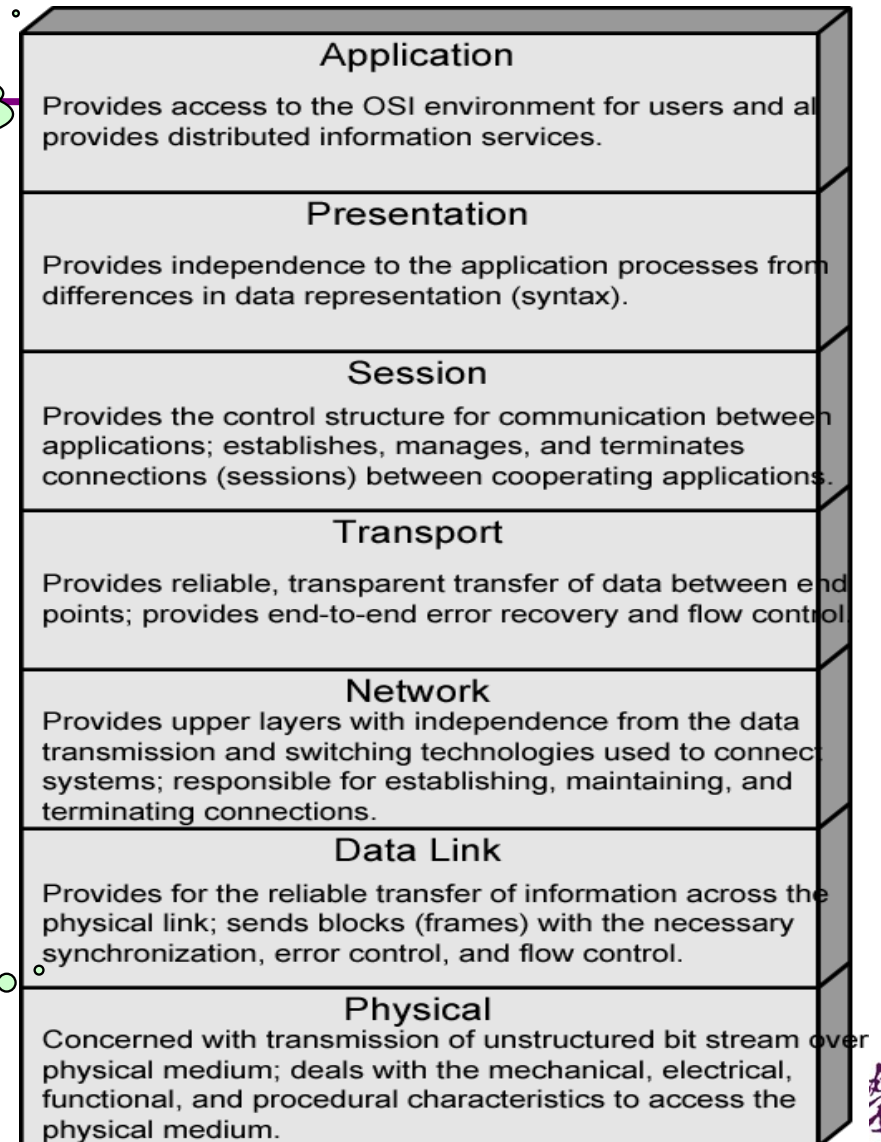
听说同步

摘机拨号

PBX 中转

信号传输

插口、双
绞线





Physical Layer



- Transfers bits across link
- Specification of the **physical aspects** of a comm link
 - **Mechanical**: cable, plugs, pins...
 - **Electrical/optical**: modulation, signal strength, voltage levels, bit times, ...
 - **Functional/procedural**: activate, maintain, deactivate physical links...
- **Physical interface** between devices
 - Ethernet, DSL, cable modem, telephone modems, ...
 - Twisted-pair cable, coaxial cable, optical fiber, radio, infrared, ...



Data Link Layer

- Groups bits into **frames**
- Activation, maintenance, & deactivation of data link **connections**
- **Transfers** frames across direct connections
- **Medium access control** for local area networks
- **Detection** of bit errors; **Retransmission** of frames
- End-to-end **flow control**
- Higher layers may assume **error free transmission**



Network Layer

- Transfers packets across **multiple links / multiple networks**
- **Addressing** must scale to large networks
- Nodes jointly execute **routing** algorithm to determine paths across the network
- **Forwarding** transfers packet across a node
- **Congestion control** to deal with traffic surges
- **Connection** setup, maintenance, and teardown when connection-based



Transport Layer

- Exchange of data **between end systems**
 - Transfers data end-to-end from process in one host to process in another host
- **Reliable** stream transfer or quick-and-simple single-block transfer
 - Error free
 - In sequence
 - No losses
 - No duplicates
- **Connection** setup, maintenance, and release



Upper Layers

- Session
 - Control of dialogues between applications
 - Dialogue discipline
 - Grouping data
 - Checkpoint recovery
- Presentation
 - Machine-independent representation of data
 - Data formats and coding
 - Data compression & encryption
- Application
 - Means for applications to access OSI environment

**Incorporated into
Application Layer Now**

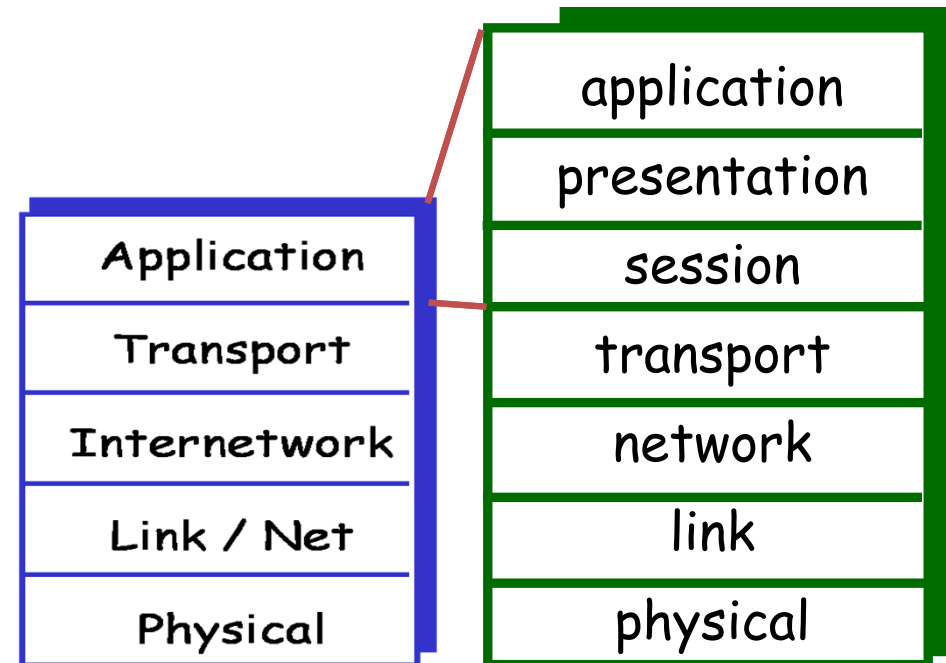


TCP/IP Protocol Architecture

Used by the global **Internet**

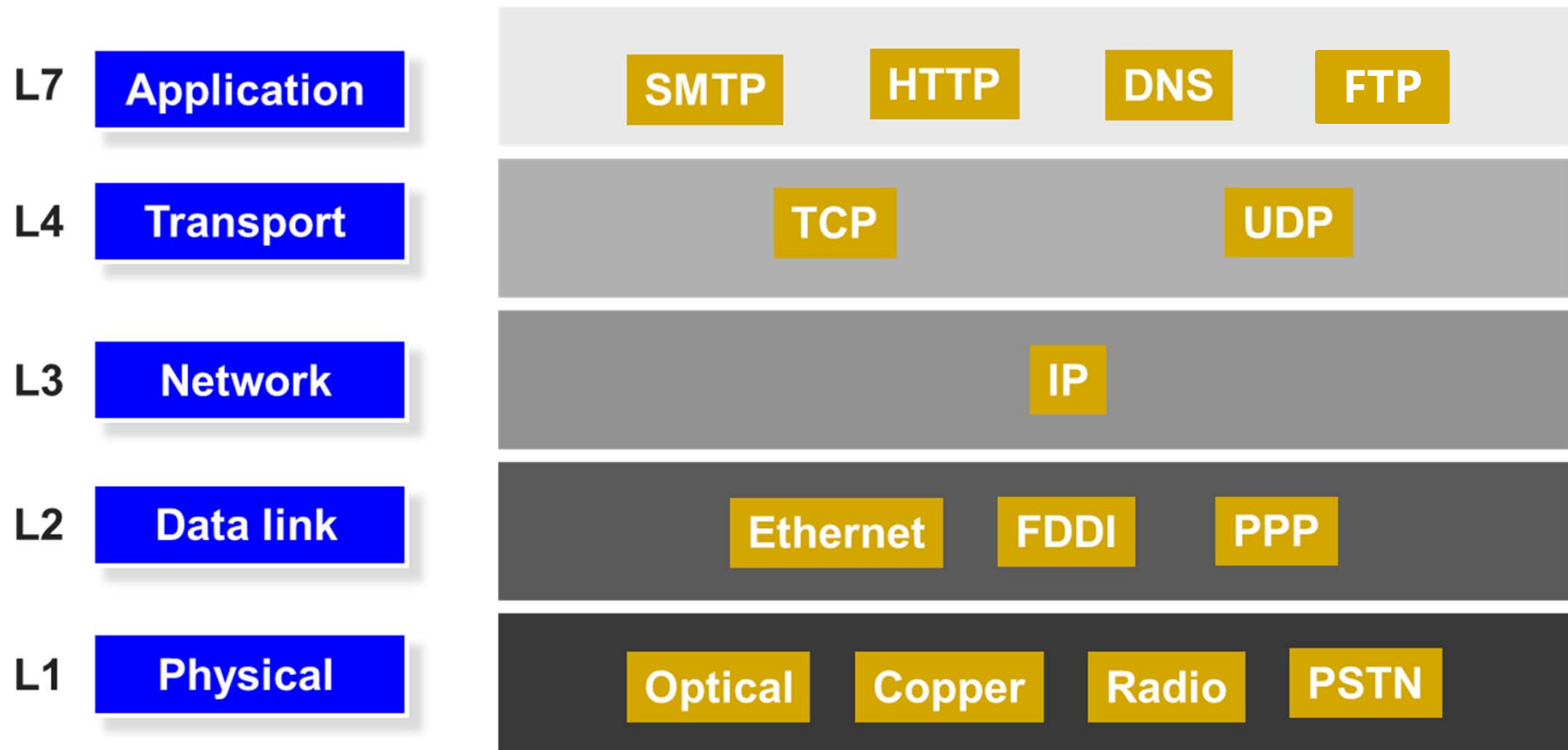
- **Application:** supporting network applications
 - FTP, SMTP, HTTP
- **Transport:** process-process data transfer
 - TCP, UDP
- **Internetwork:** routing of datagrams across net of nets
 - IP, routing protocols
- **Link:** data transfer between neighboring routers / hosts
 - PPP, Ethernet
- **Physical:** bits "on the wire"

TCP/IP protocol stack vs. OSI



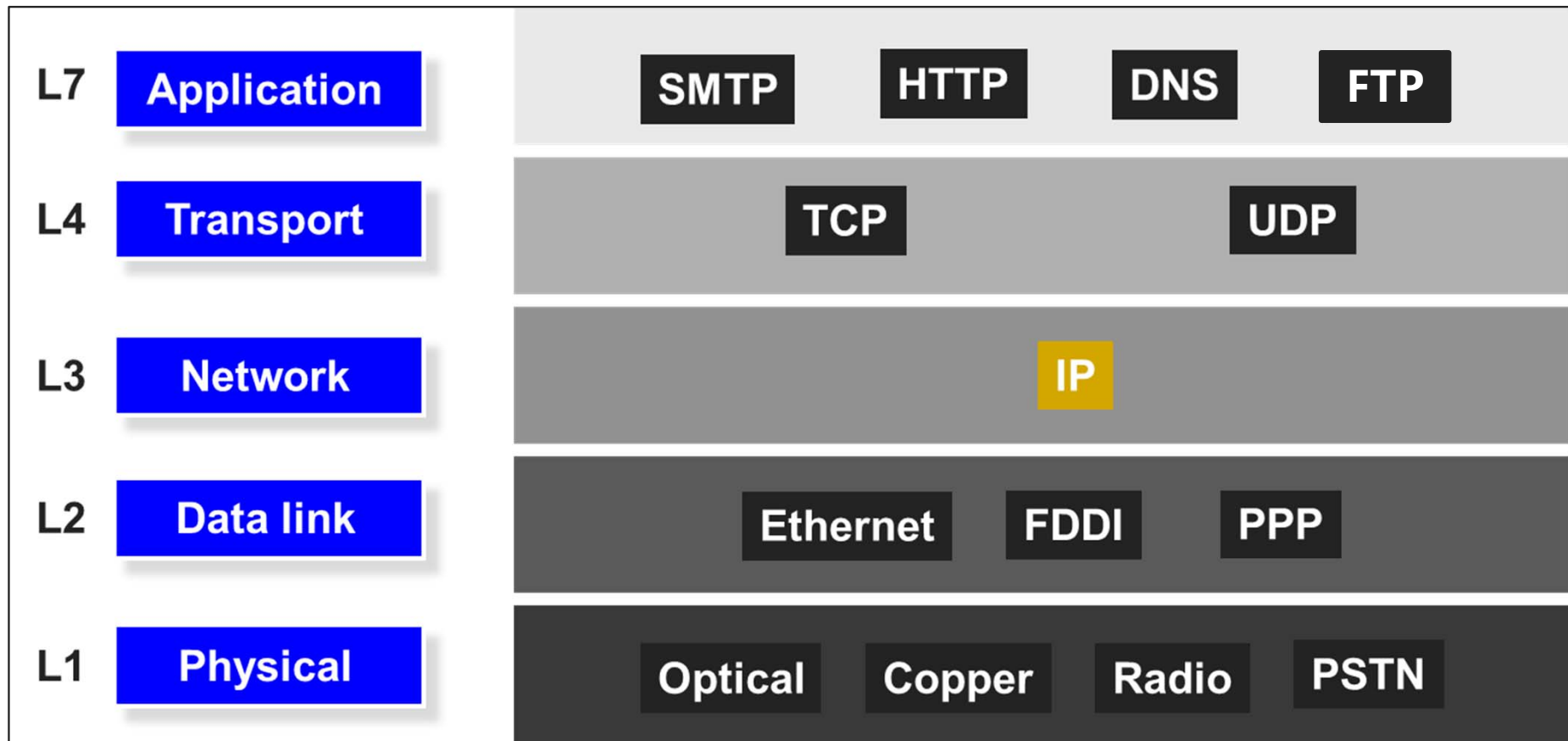


Protocols at different layers



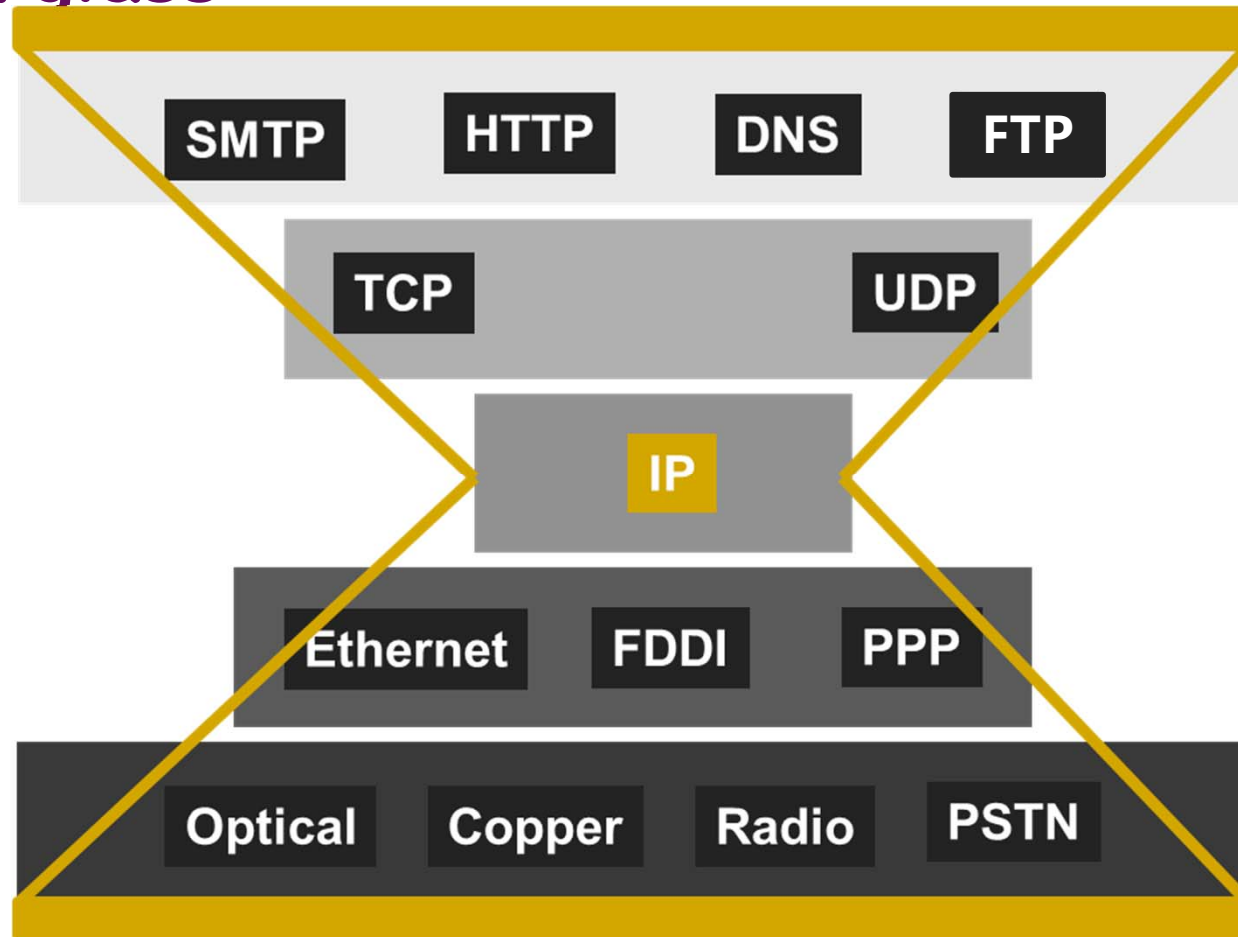


ONE network layer protocol





IP is the narrow waist of the layering hourglass



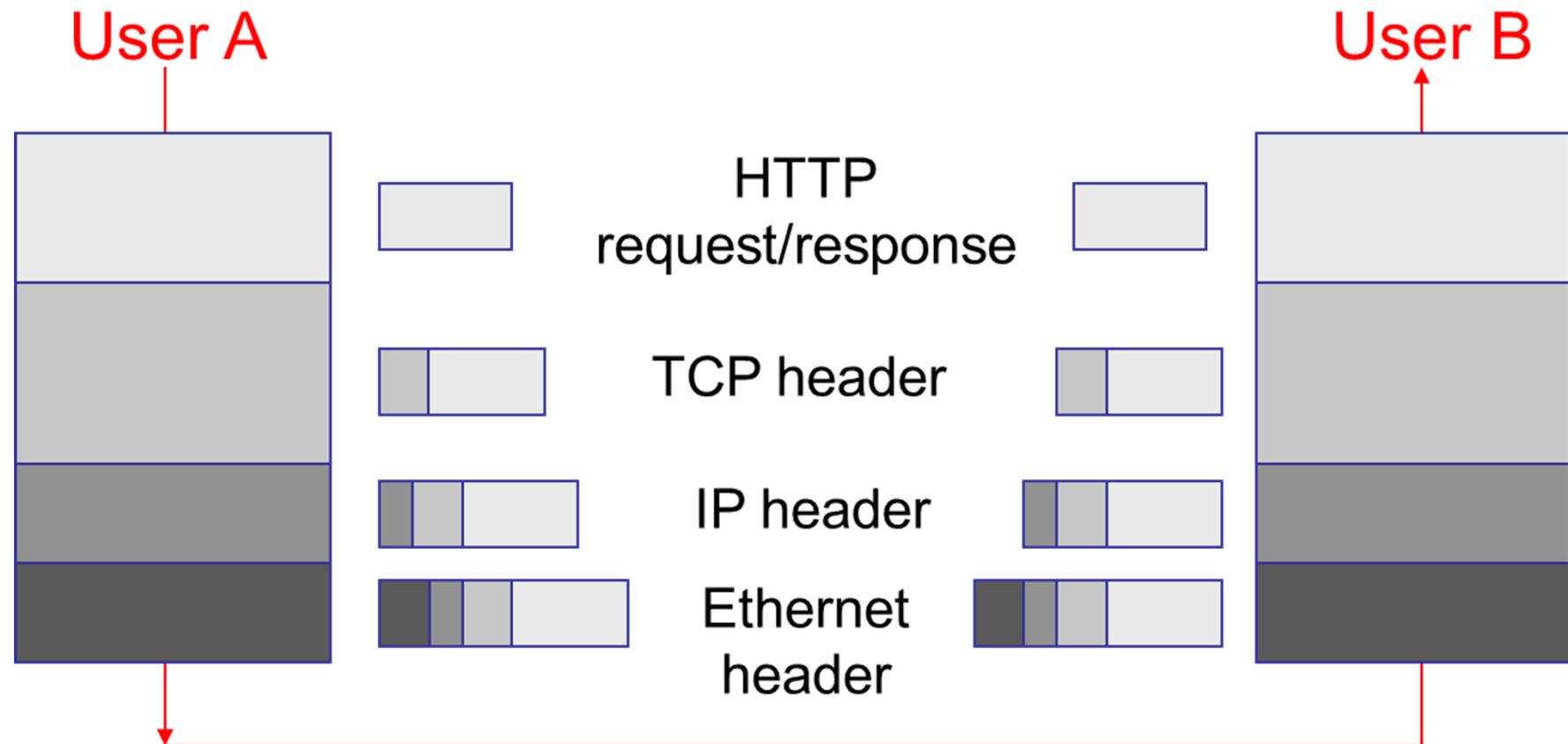


Implications of hourglass

- Single network-layer protocol (IP)
- Allows arbitrary networks to interoperate
 - Any network that supports IP can exchange packets
- **Decouples** applications from low-level networking technologies
 - Applications function on all networks
- Supports simultaneous innovations above and below IP
- But changing IP itself is hard (e.g., IPv4 → IPv6)



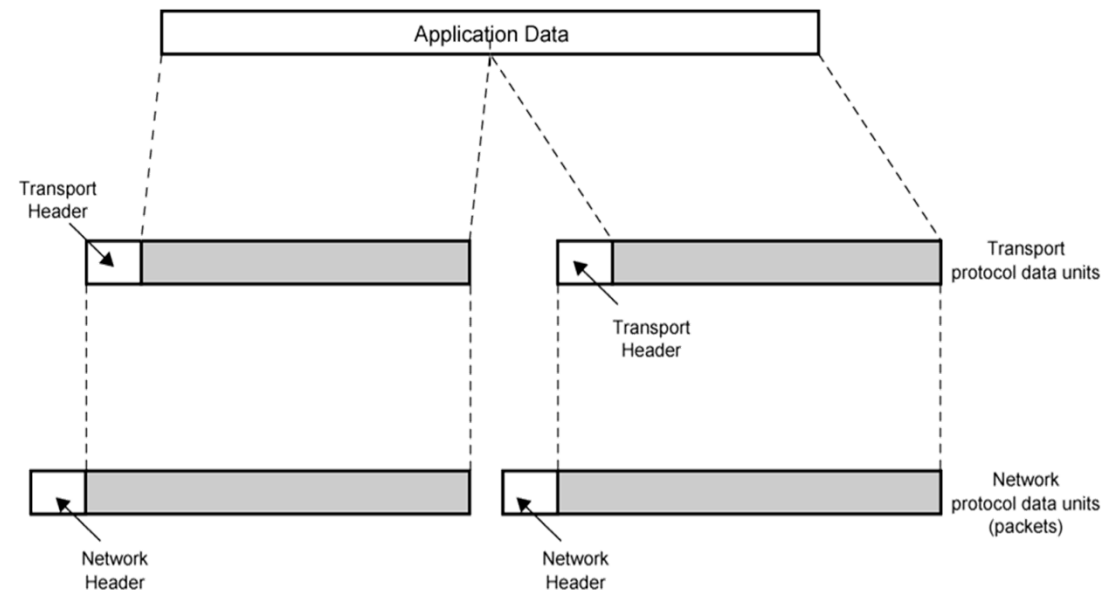
Layer encapsulation: Protocol headers





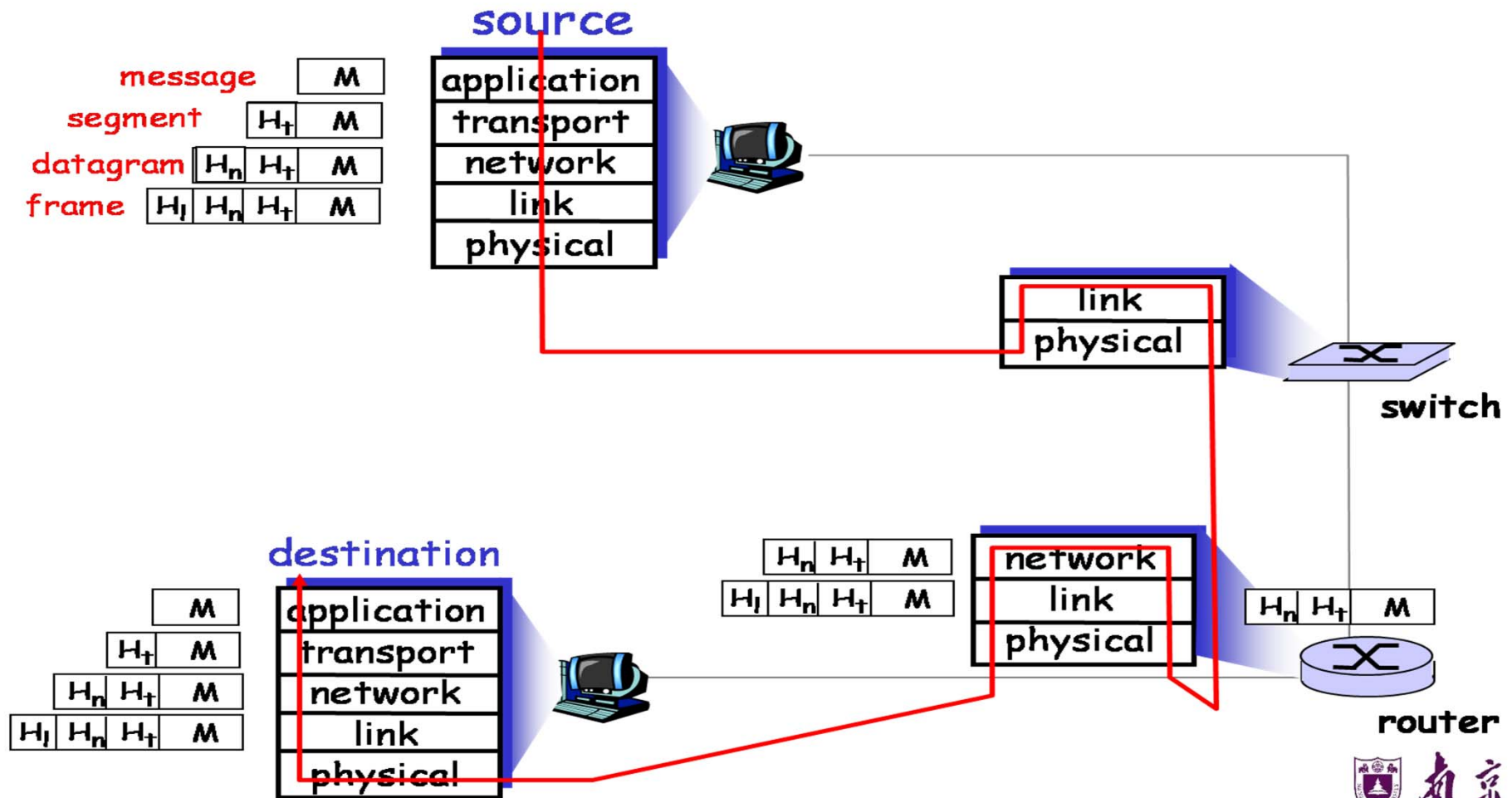
Protocol Data Units

- At each layer, **Control info** is added to **user data** to ease communication, e.g.
- Transport layer segments application data
- Each segment has a **transport header** added
 - Destination port
 - Sequence number
 - Error detection code
- This gives a **transport** protocol data unit (PDU)





Encapsulation





Pros and cons of layering

- Why layers?
 - Reduce complexity
 - Improve flexibility
- Why not?
 - Higher overheads
 - Cross-layer information often useful



Computer Networks and Internet

- Basic Concepts and Questions
- Internet History
- Protocol Layers and Service Model
- Network Performance



Performance metrics

- Delay
- Loss
- Throughput



Delay

- How long does it take to send a packet from its source to destination?



Delay

- Consists of four components
 - Transmission delay
 - Propagation delay
 - Queuing delay
 - Processing delay
- due to link properties
- due to traffic mix and switch internals

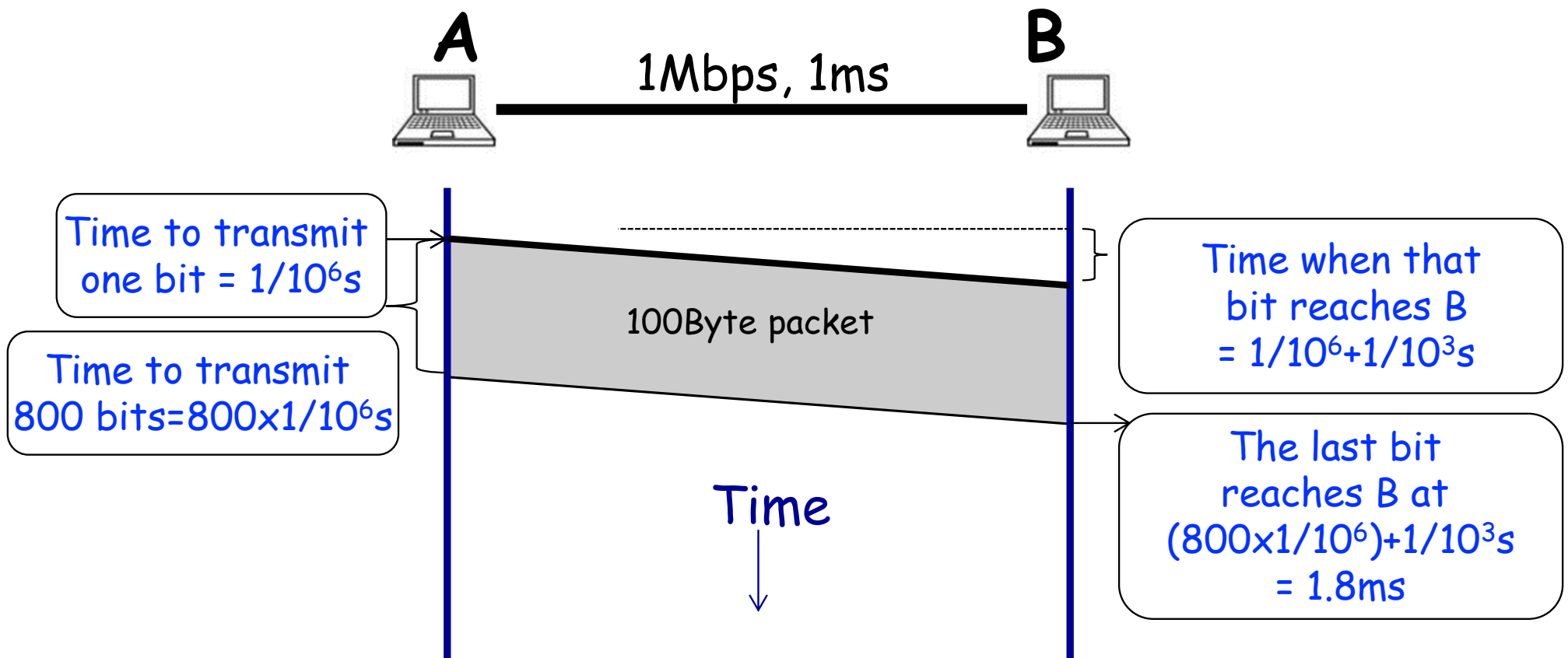


A network link

- Transmission delay (传输时延)
 - How long does it take to push all the bits of a packet into a link?
 - Packet size / Transmission rate of the link
 - e.g., 1000 bits / 100 Mbits per sec = 10^{-5} sec
- Propagation delay (传播时延)
 - How long does it take to move one bit from one end of a link to the other?
 - Link length / Propagation speed of link
 - E.g., 30 kilometers / 3×10^8 meters per sec = 10^{-4} sec

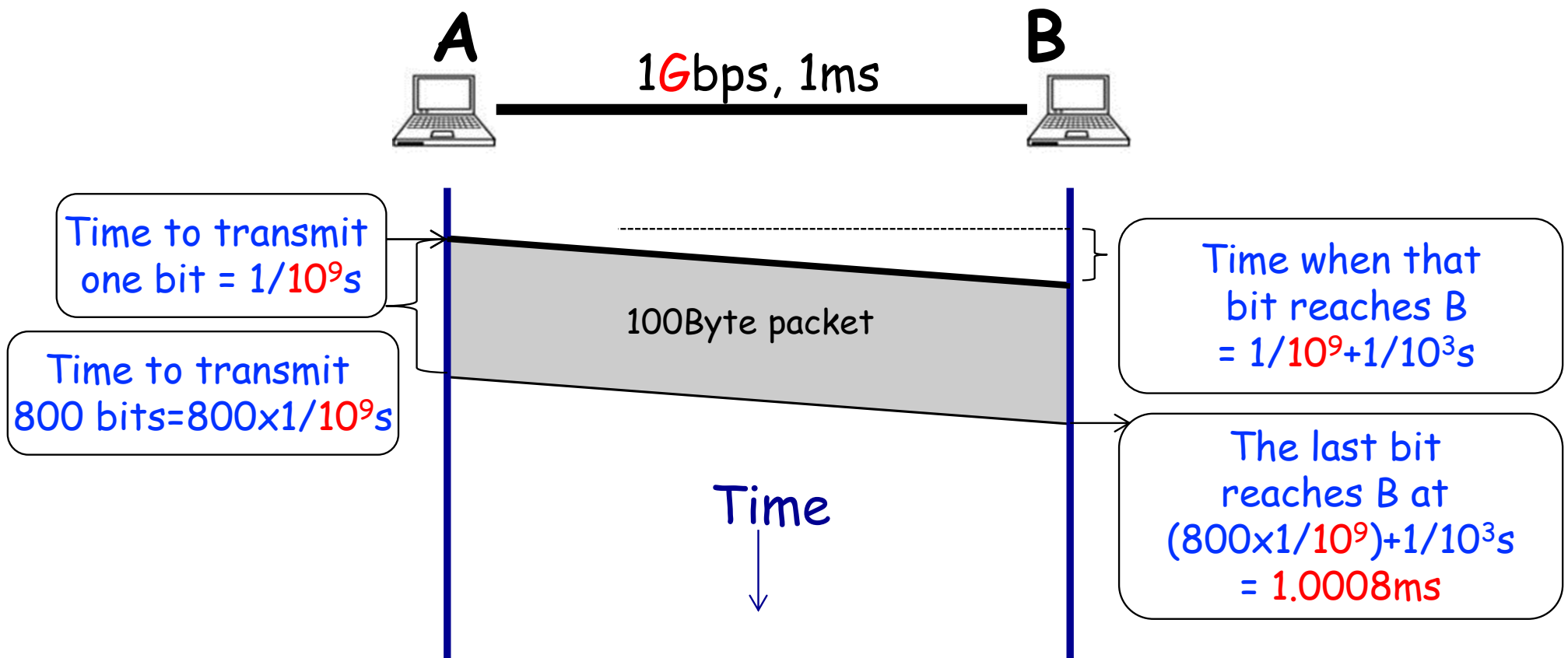


Packet delay Sending a 100-byte packet



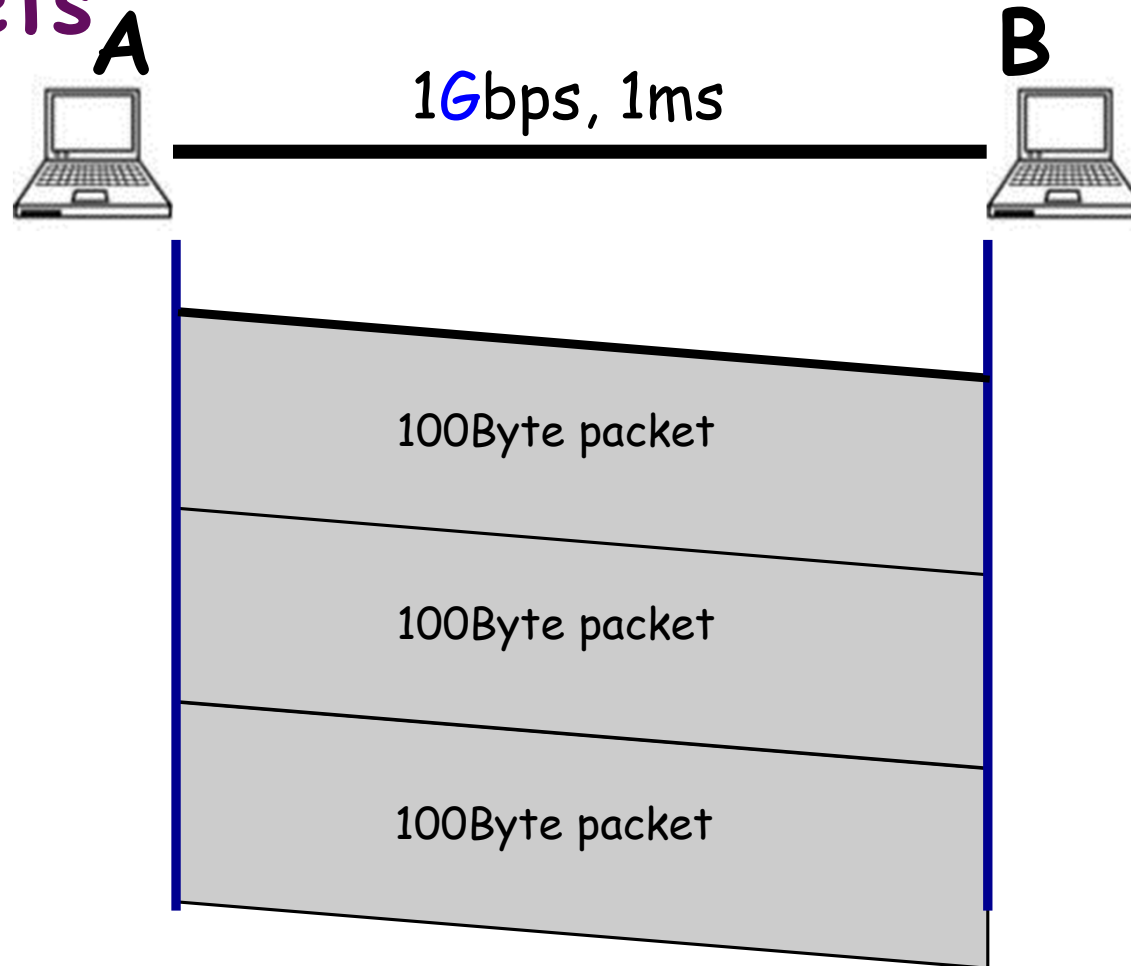


Packet delay Sending a 100-byte packet



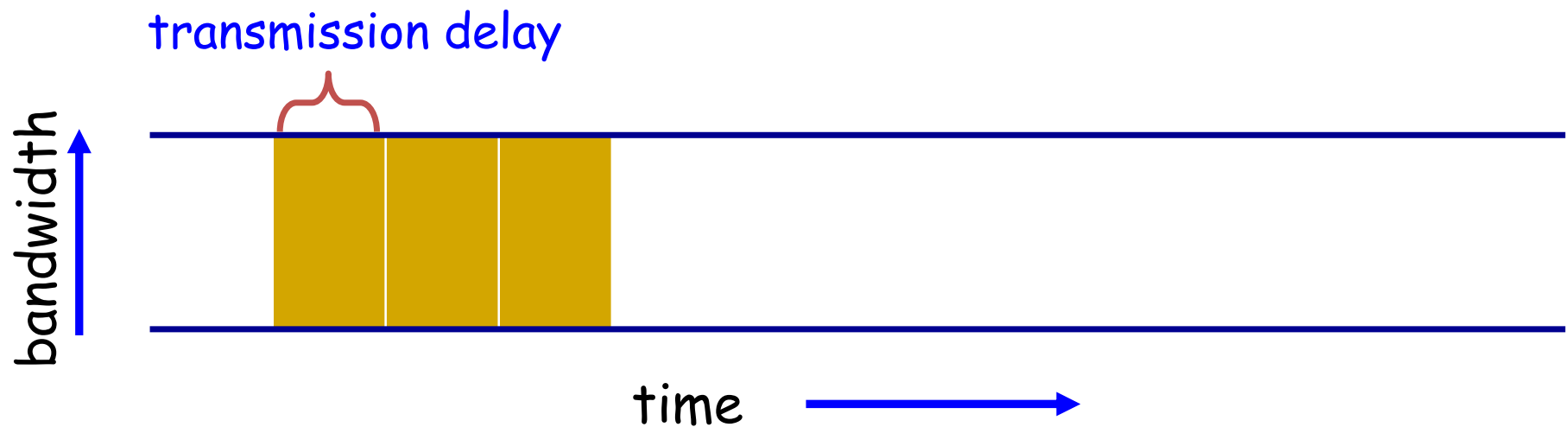


Sending a large file using 100-byte packets





Pipe view of a link



- Transmission delay decreases as bandwidth increases

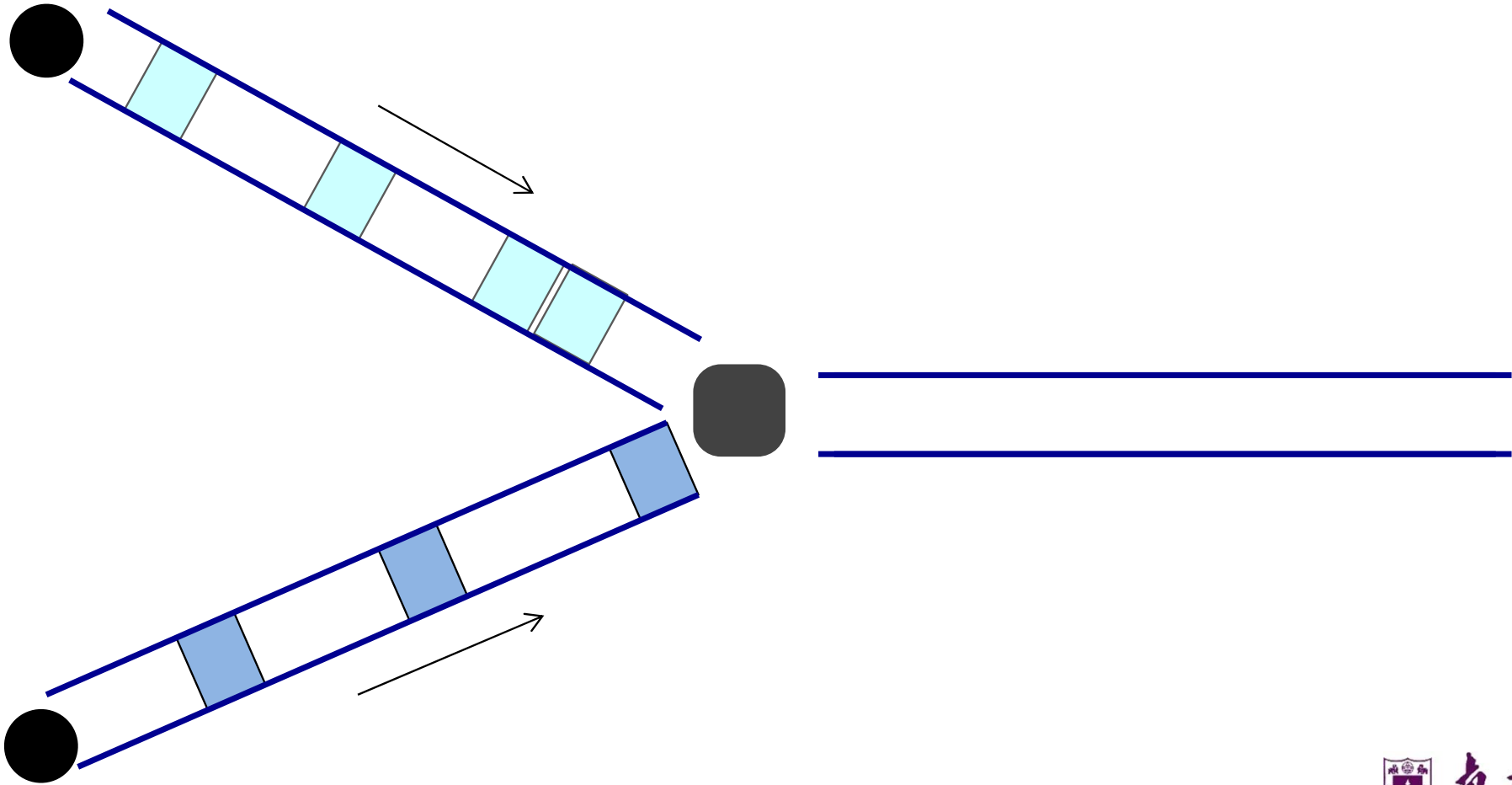


Queuing delay

- How long does a packet have to sit in a buffer before it is processed?

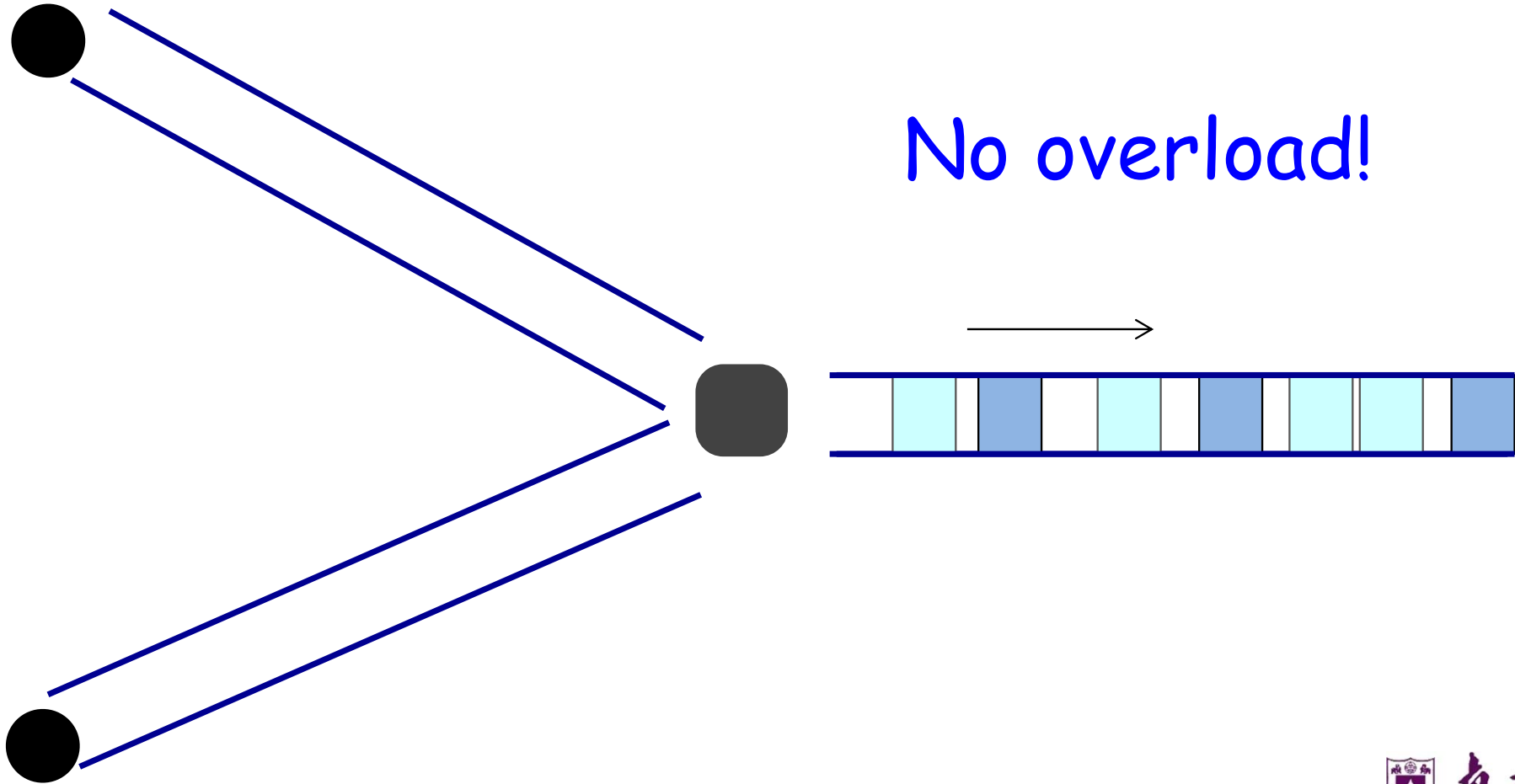


Queueing delay: "pipe" view



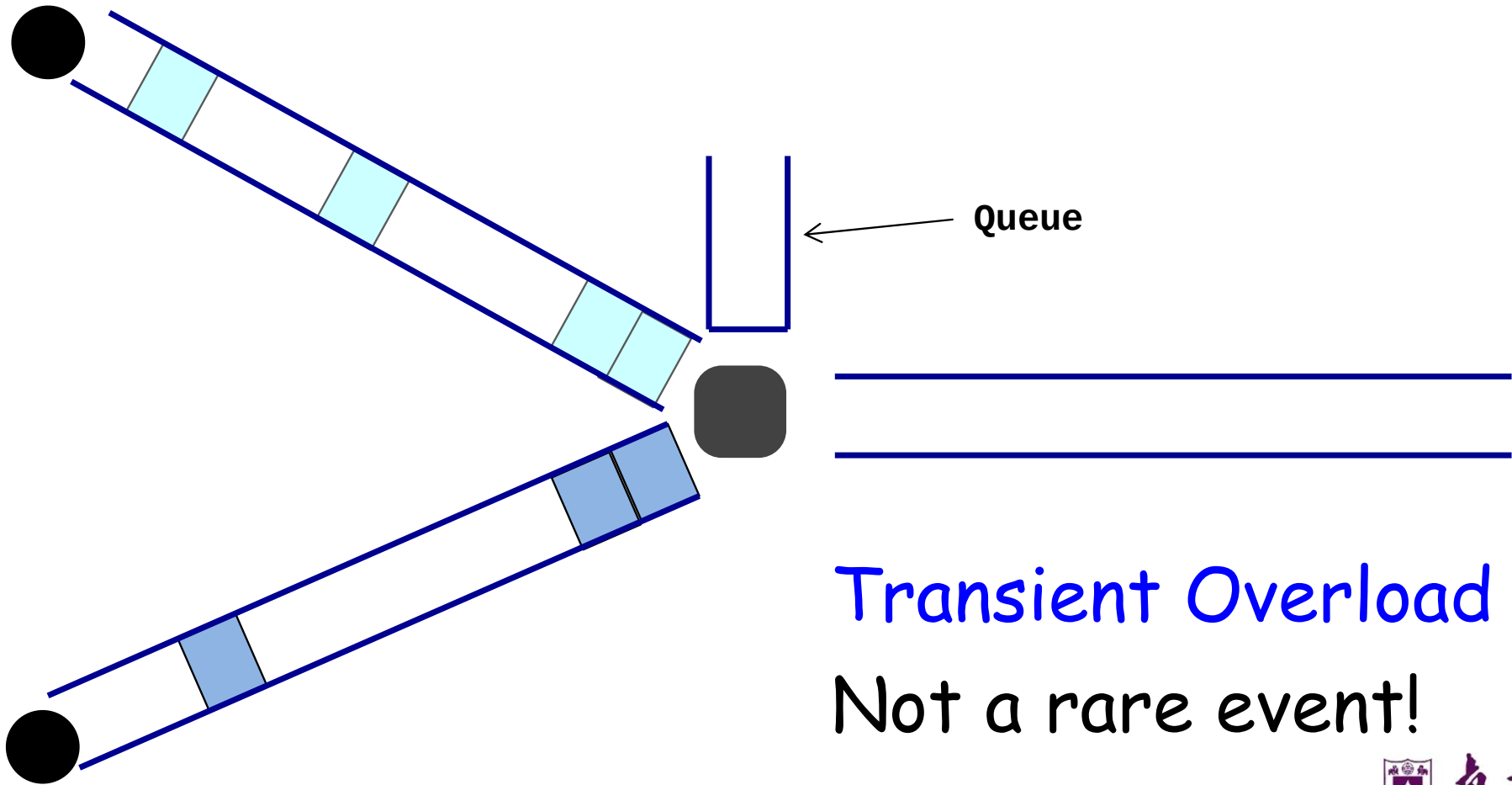


Queueing delay: "pipe" view



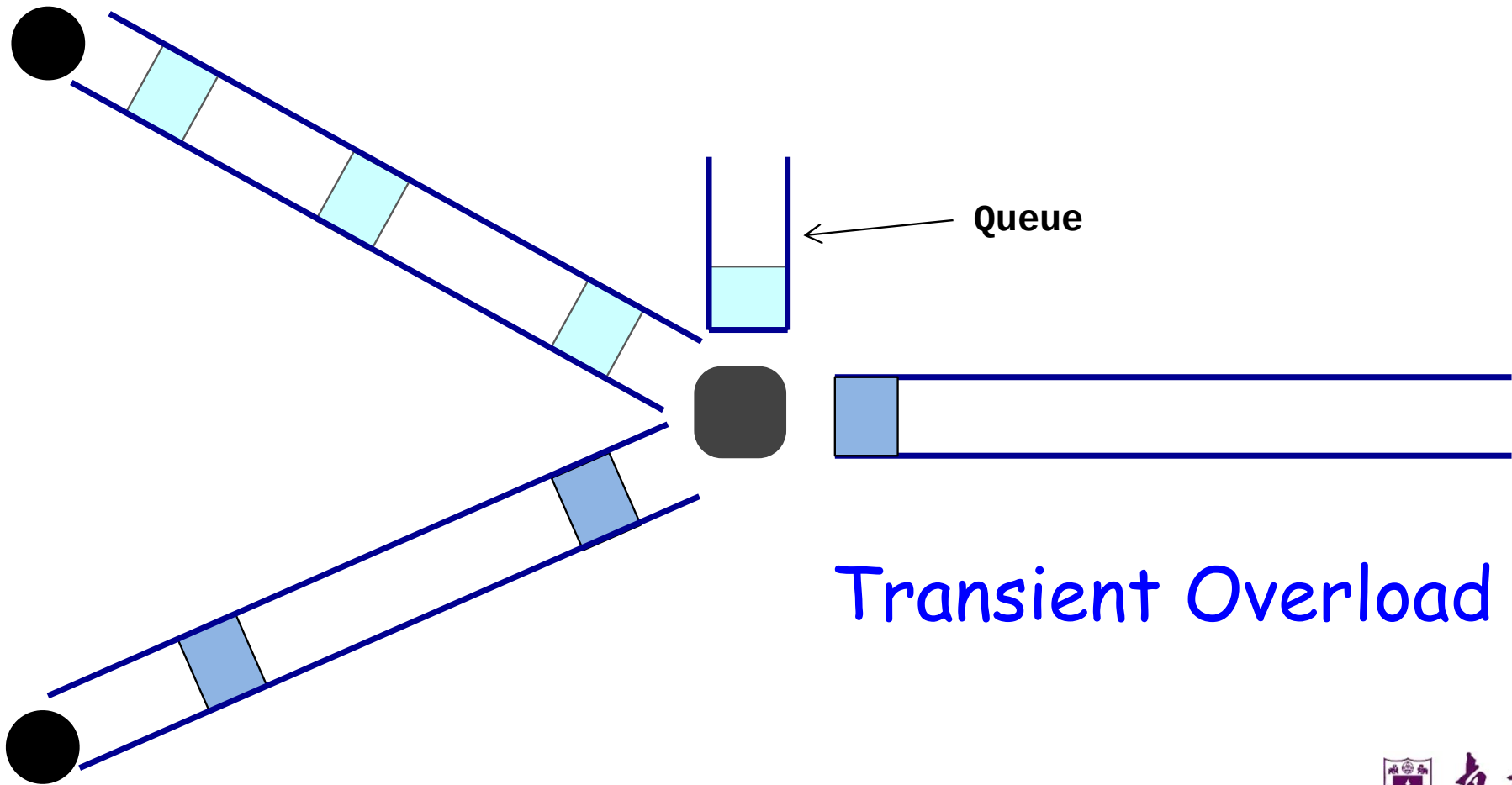


Queueing delay: “pipe” view



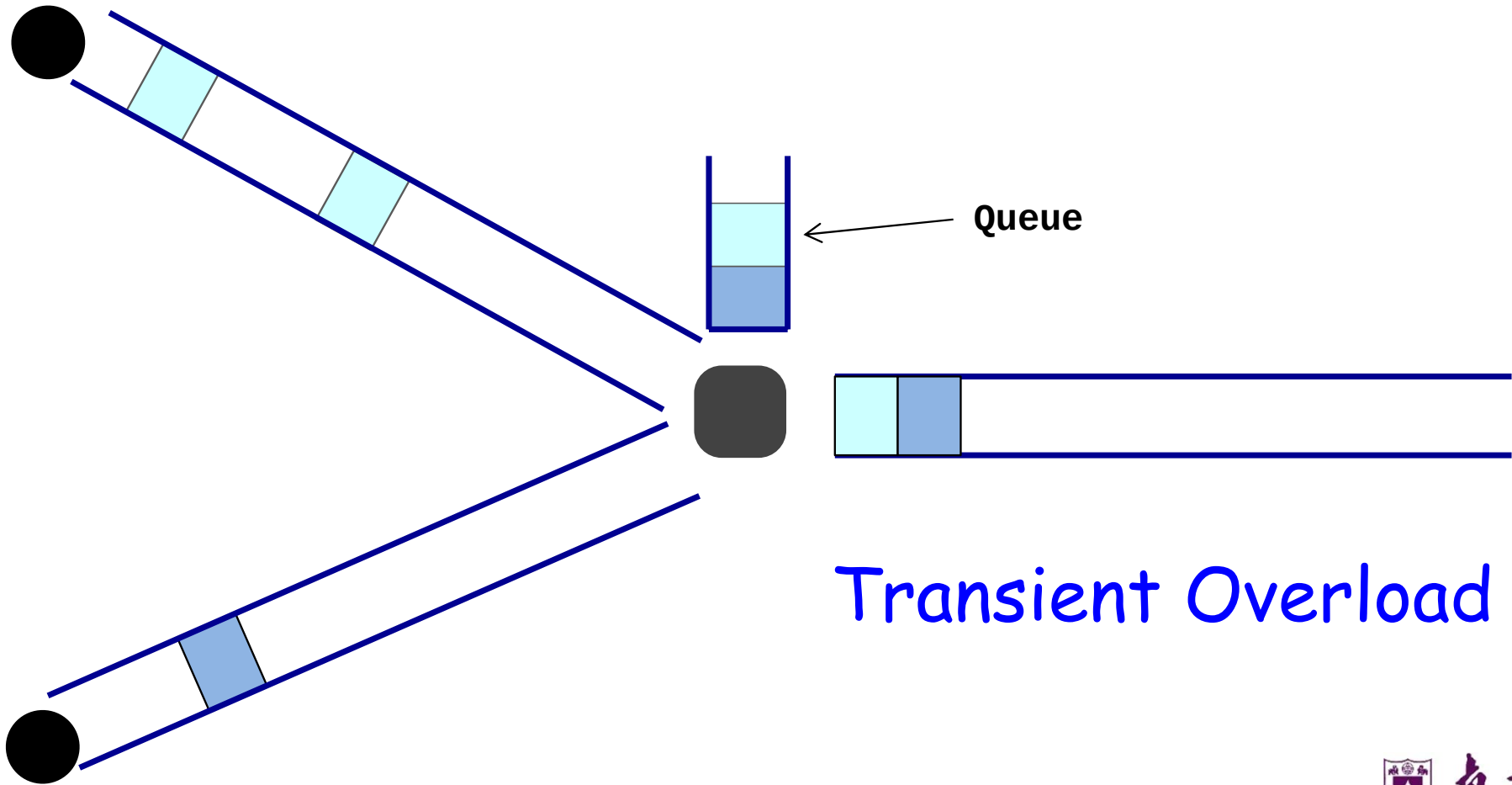


Queueing delay: "pipe" view



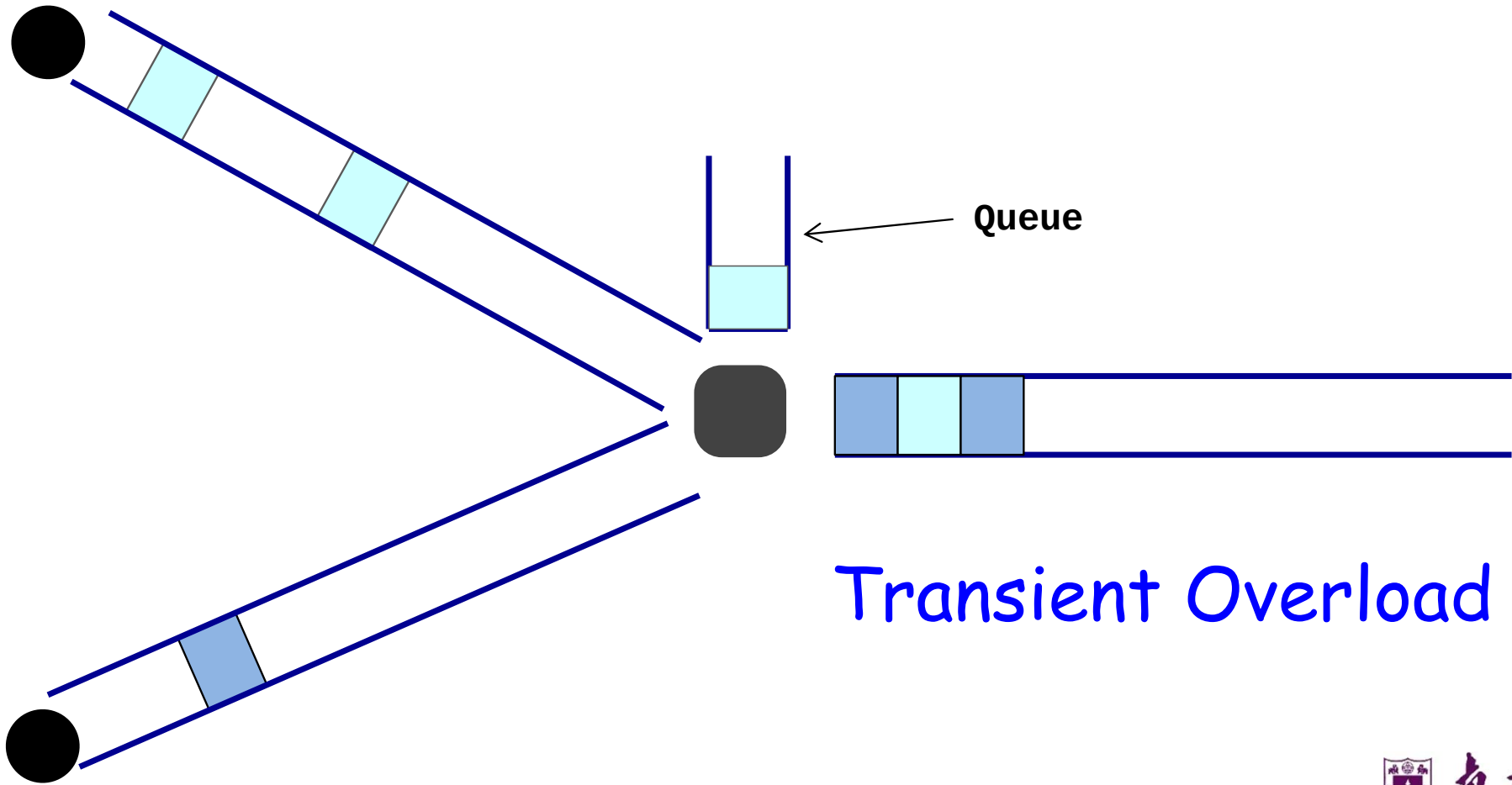


Queueing delay: "pipe" view





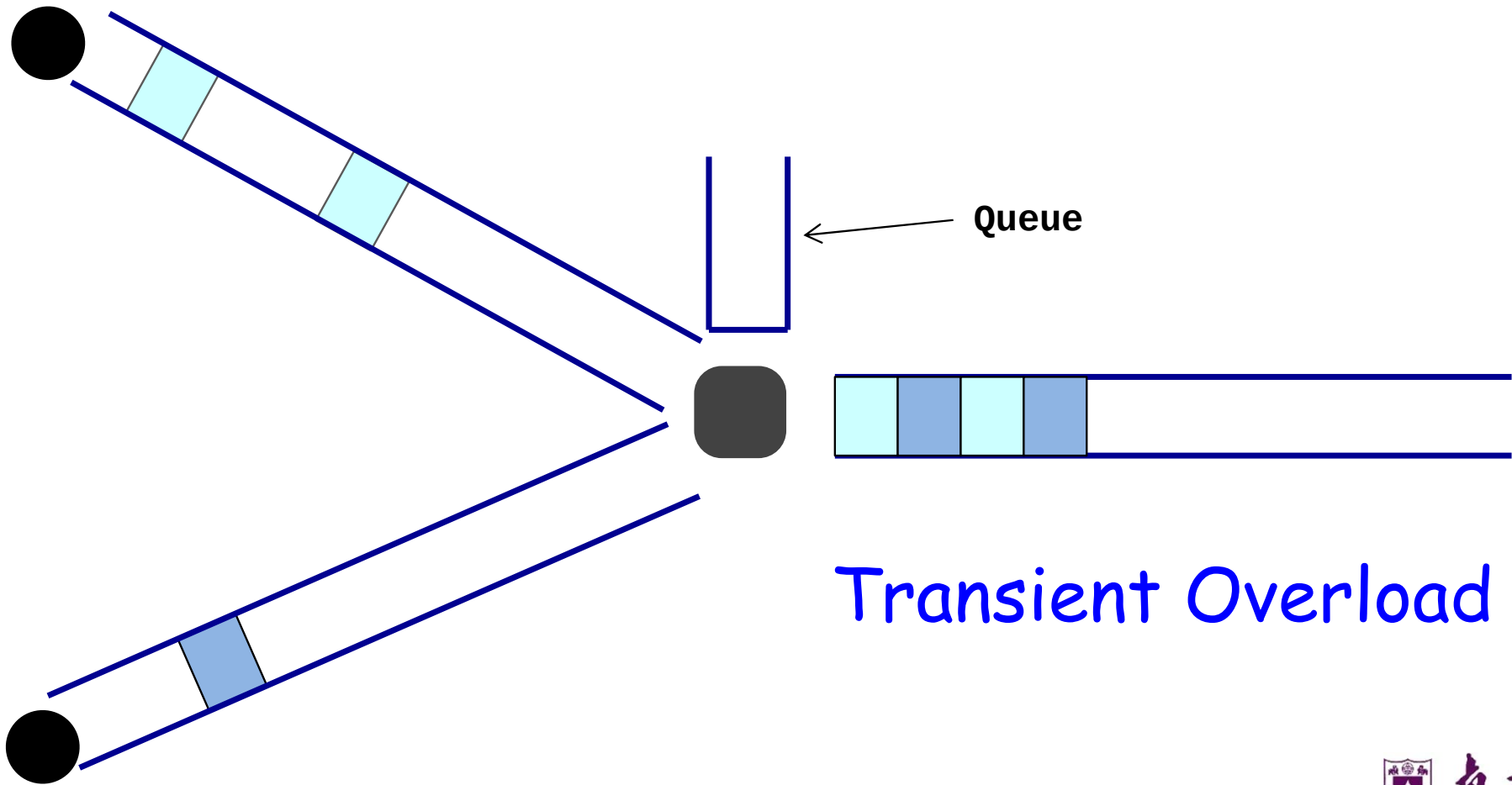
Queueing delay: "pipe" view



Transient Overload

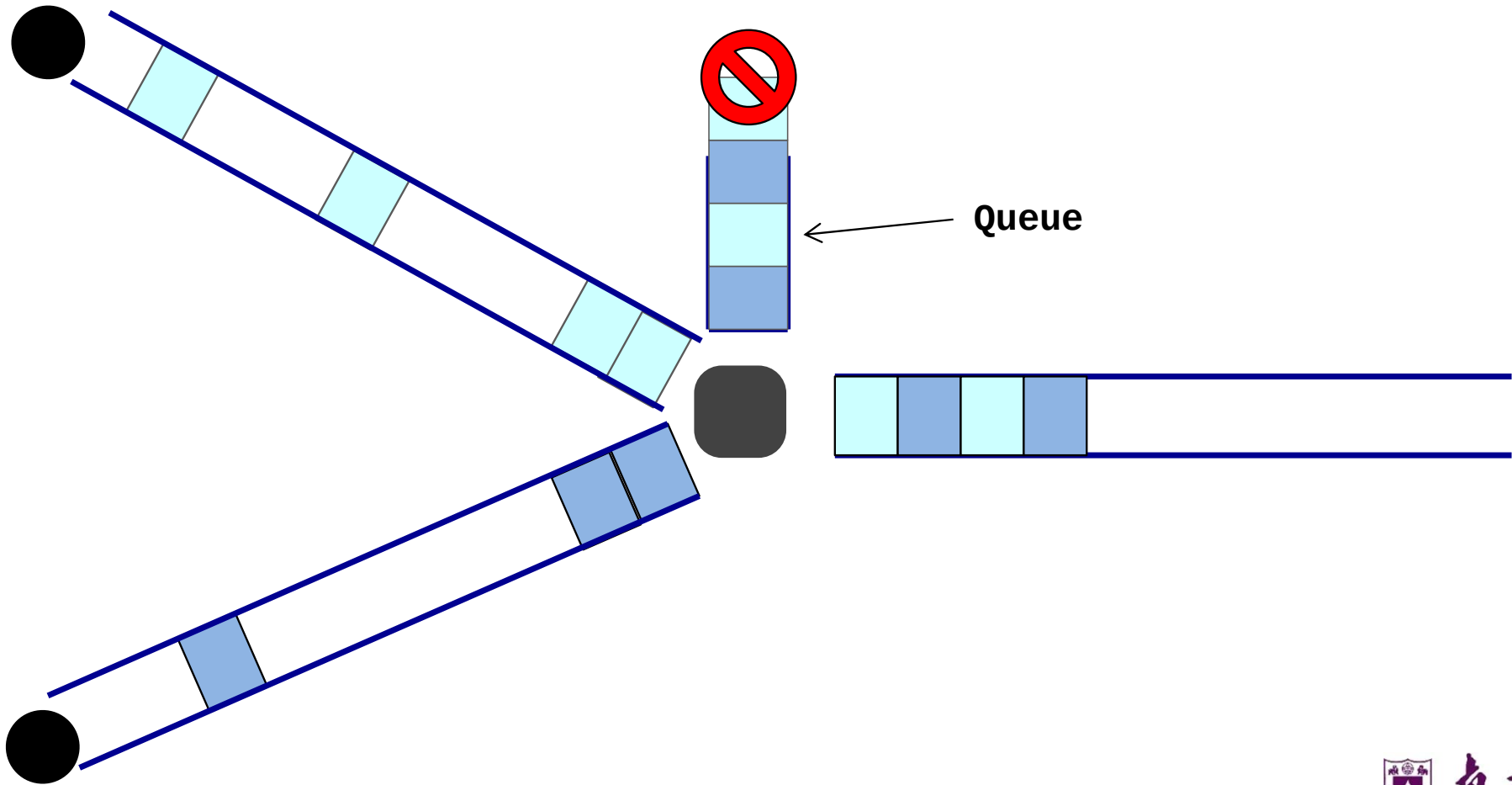


Queueing delay: “pipe” view





Queueing delay: "pipe" view





Queueing delay

- How long does a packet have to sit in a buffer before it is processed?
- Depends on traffic pattern
 - Arrival rate at the queue
 - Nature of arriving traffic (bursty or not?)
 - Transmission rate of outgoing link



Queueing delay

- How long does a packet have to sit in a buffer before it is processed?
- Characterized with statistical measures
 - Average queuing delay
 - Variance of queuing delay
 - Probability delay exceeds a threshold value



Basic queueing theory terminology

- Arrival process: how packets arrive
 - Average rate A
 - Peak rate P
- W : average time packets wait in the queue
 - W for "waiting time"
- L : average number of packets waiting in the queue
 - L for "length of queue"



Little's Law (1961)

- $L = A \times W$
- Compute L : count packets in queue every second
 - How often does a single packet get counted? W times
- Why do you care?
 - Easy to compute L , harder to compute W

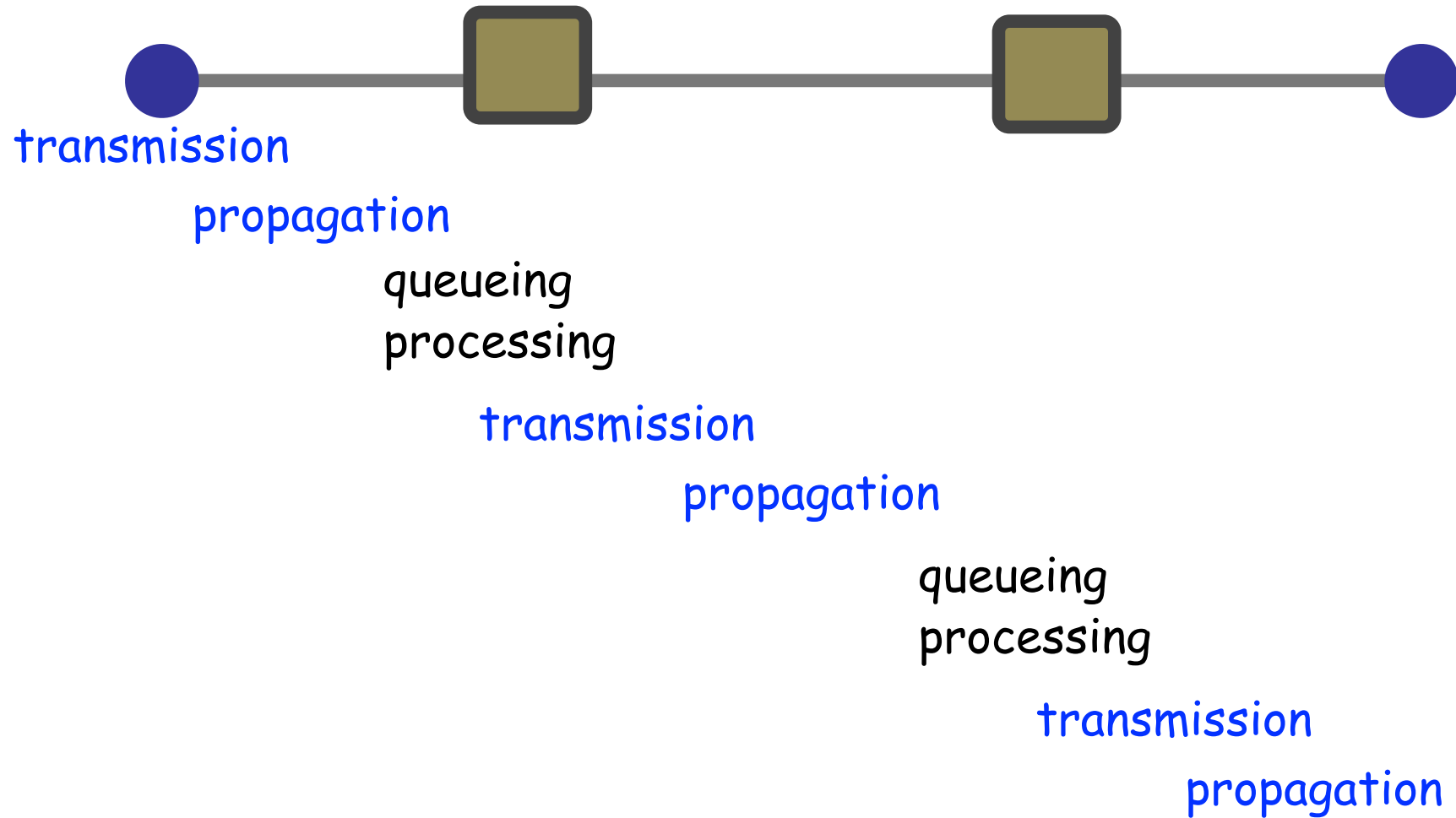


Processing Delay

- How long does the switch take to process a packet?
 - Negligible



End-to-end delay





Loss

- What fraction of the packets sent to a destination are dropped?



Throughput

- At what rate is the destination receiving data from the source



Throughput

Transmission rate R bits/sec



File of size F bits

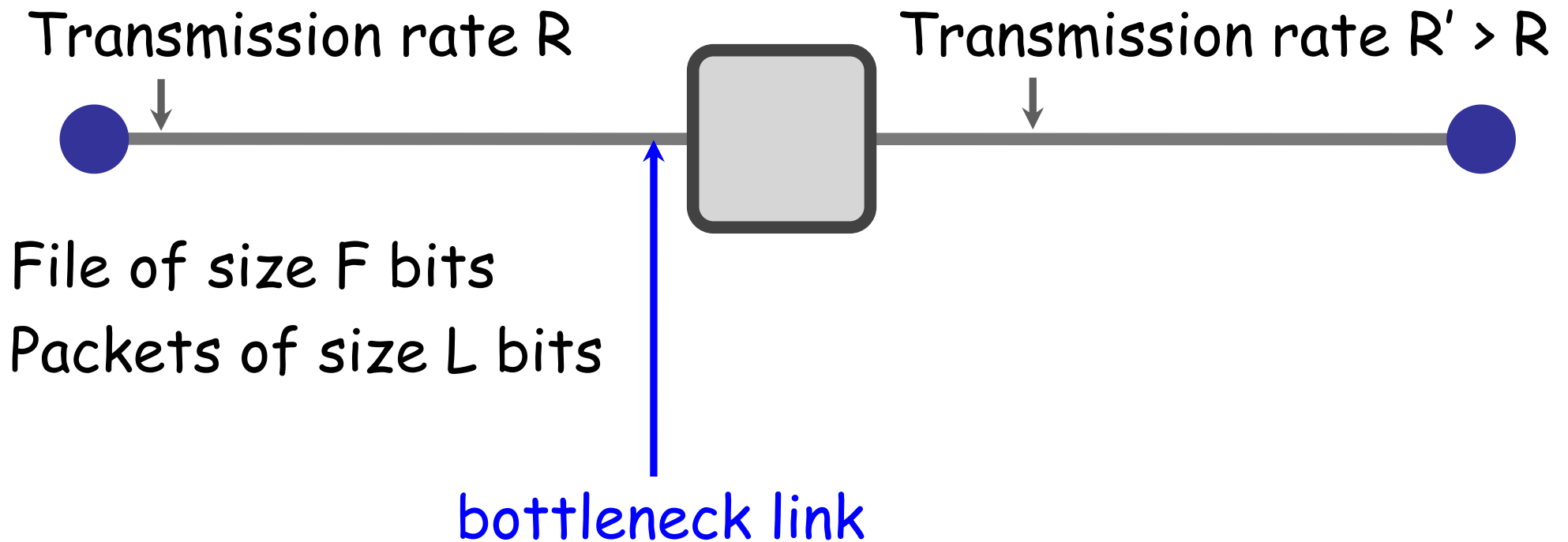
Packets of size L bits

Transfer time $(T) = F/R$ + propagation delay

Average throughput = $F/T \approx R$



End-to-end throughput



$$\text{Average throughput} = \min\{R, R'\} = R$$



提问

Q & A